

Data Communications and Networking (BCS502 Module 1)

1. Data Communications

Definition: Data communications is the process of exchanging data between two devices through a transmission medium, such as a wire cable or wireless signals. It forms the backbone of modern telecommunication systems, enabling the sharing of information like text, images, or video across local or remote locations. This process is integral to systems like the Internet, telephone networks, and computer networks, requiring both hardware (e.g., cables, routers) and software (e.g., protocols, applications) to function effectively.

Fundamental Characteristics: Data communication systems must meet four key requirements to be effective:

1. **Delivery:** Data must reach the intended recipient accurately. For example, an email sent to a specific address must arrive only at that address, not to unintended recipients. This ensures privacy and reliability in communication.
2. **Accuracy:** Transmitted data must be free of errors. If data is altered during transmission (e.g., due to noise or interference) and not corrected, it becomes unusable. For instance, a corrupted file download is rendered useless if errors are not fixed.
3. **Timeliness:** Data must be delivered promptly to be useful, especially in real-time applications like video conferencing or live streaming. Late data can disrupt user experience, such as causing buffering in a video call. Real-time transmission ensures data is delivered as produced, in order, with minimal delay.
4. **Jitter:** Refers to the variation in packet arrival times, which affects the quality of audio or video streams. For example, if video packets arrive unevenly (some at 30 ms, others at 40 ms), the result is choppy playback. Low jitter is critical for smooth multimedia experiences.

Components of a Data Communications System (Figure 1.1): A data communications system comprises five essential components, each playing a distinct role:

1. **Message:** The information to be shared, which can take various forms like text (e.g., emails), numbers (e.g., financial data), images (e.g., photos), audio (e.g., voice calls), or video (e.g., streaming movies). The versatility of message types reflects the diverse applications of data communications.
2. **Sender:** The device that initiates the data transmission, such as a computer sending an email, a smartphone making a call, or a video camera streaming footage. The sender encodes the message into a format suitable for transmission.
3. **Receiver:** The device that receives and decodes the message, such as a computer displaying a webpage or a TV showing a broadcast. The receiver must be compatible with the sender's format to interpret the data correctly.

4. **Transmission Medium:** The physical path through which data travels, such as wired media (twisted-pair, coaxial, or fiber-optic cables) or wireless media (radio waves, microwaves). The choice of medium impacts speed, reliability, and distance of communication.
5. **Protocol:** A set of rules ensuring effective communication between devices. Protocols define how data is formatted, transmitted, and received. For example, without a common protocol, two devices might be connected but unable to communicate, like speakers of different languages.

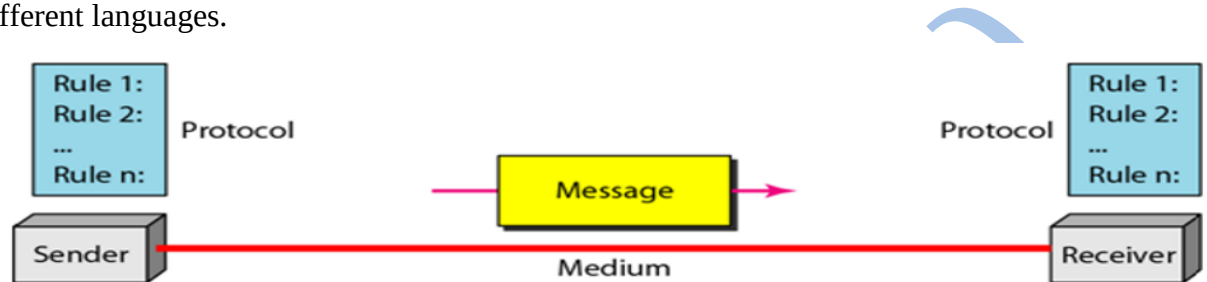


Figure 1.1 *Components of a data communication system*

Data Representation: Data is represented in various forms to suit different communication needs:

- **Text:** Represented as bit patterns (sequences of 0s and 1s). The **Unicode** standard uses 32 bits to encode characters from any language, ensuring global compatibility. The **ASCII** code, a subset of Unicode, uses 7 bits for 127 characters (Basic Latin). For example, the letter 'A' in ASCII is 01000001.
- **Numbers:** Converted directly to binary (e.g., decimal 5 becomes binary 101) for efficient mathematical processing, bypassing text-based codes like ASCII to simplify operations.
- **Images:** Represented as a matrix of pixels, where each pixel is a small dot assigned a bit pattern. For black-and-white images, a 1-bit pattern (0 or 1) suffices. Grayscale images use multiple bits (e.g., 2 bits for four shades: 00 for black, 01 for dark gray, 10 for light gray, 11 for white). Color images use models like **RGB** (red, green, blue) or **YCM** (yellow, cyan, magenta), where each color component is assigned a bit pattern based on intensity.
- **Audio:** Continuous signals from sound or music, unlike discrete text or numbers. For example, a microphone converts voice into a continuous electrical signal, which is then digitized for transmission.
- **Video:** Can be continuous (e.g., live TV broadcasts) or a sequence of discrete images (e.g., animated GIFs) that create the illusion of motion. Video requires high bandwidth due to its data-intensive nature.

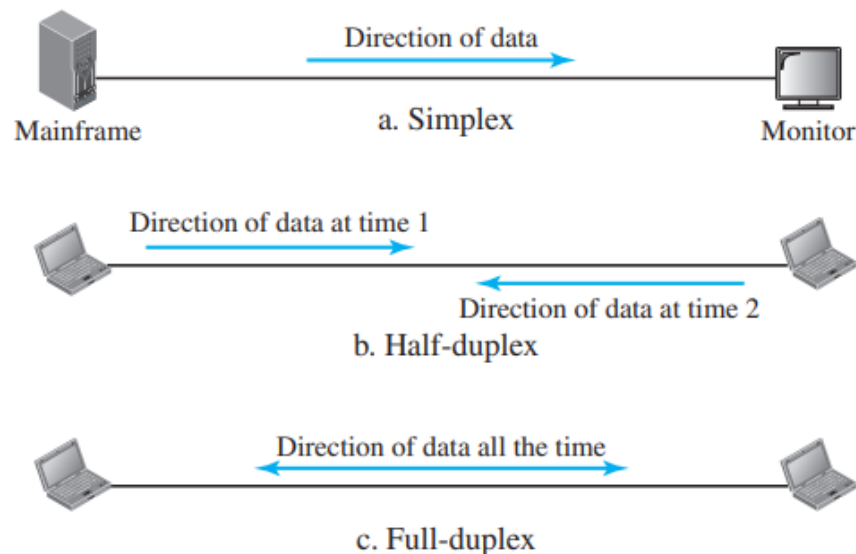
Data Flow Modes (Figure 1.2): Data flow defines how devices communicate:

1. **Simplex:** Unidirectional communication, like a one-way street. Only one device transmits, and the other receives (e.g., a keyboard sending input to a computer, or a

monitor displaying output). It uses the full channel capacity for one direction, maximizing efficiency for specific applications.

2. **Half-Duplex:** Bidirectional but not simultaneous, like a single-lane road where traffic alternates. Devices take turns transmitting (e.g., walkie-talkies or CB radios). The entire channel capacity is used by the transmitting device, making it efficient for non-simultaneous communication.
3. **Full-Duplex:** Simultaneous bidirectional communication, like a two-way street. Both devices can send and receive at the same time (e.g., a telephone call where both parties talk and listen simultaneously). The channel capacity is split via separate paths or time division, enabling continuous interaction.

Figure 1.2 *Data flow (simplex, half-duplex, and full-duplex)*



Additional Explanation: Data communications form the foundation of modern networking, enabling everything from simple text messages to complex video streams. The characteristics (delivery, accuracy, timeliness, jitter) ensure reliable and efficient communication, particularly for real-time applications where delays or errors can significantly degrade user experience. The components work together seamlessly: the sender encodes the message, the medium carries it, the receiver decodes it, and protocols ensure compatibility. Data representation is critical for handling diverse data types, with standards like Unicode ensuring global interoperability. Understanding data flow modes helps in designing systems for specific needs, such as simplex for broadcasting or full-duplex for interactive applications like VoIP.

2. Networks

Definition: A network is a set of interconnected devices (e.g., computers, phones, routers) capable of communication, using wired or wireless media. Networks enable resource sharing, data exchange, and connectivity across various scales, from small home setups to global systems like the Internet.

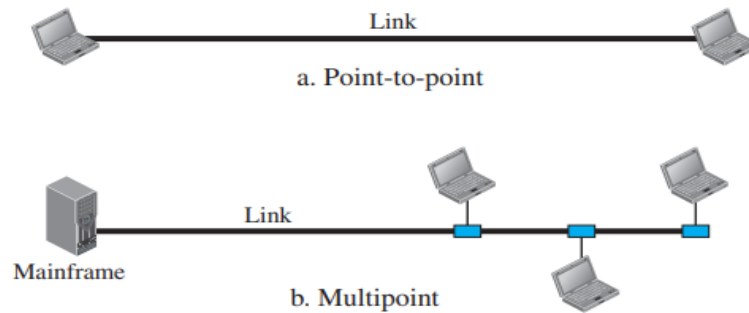
Network Criteria: Networks are evaluated based on three critical factors:

1. **Performance:** Measured by:
 - **Transit Time:** Time for a message to travel from source to destination.
 - **Response Time:** Time between sending a request and receiving a response.
 - Performance depends on factors like the number of users, transmission medium, hardware capabilities, and software efficiency. Metrics include:
 - **Throughput:** The rate of successful data transfer (e.g., Mbps). Higher throughput is desirable but can lead to congestion.
 - **Delay:** Time taken due to processing, queuing, or congestion. High throughput often increases delay, requiring careful network design to balance both.
2. **Reliability:** Assessed by failure frequency, recovery time, and resilience to catastrophic events. A reliable network minimizes downtime and quickly recovers from failures, ensuring continuous operation.
3. **Security:** Involves protecting data from unauthorized access, ensuring data integrity, and implementing recovery mechanisms for breaches. Security is critical in networks handling sensitive information, such as financial or personal data.

Physical Structures:

- **Type of Connection:**
 - **Point-to-Point:** A dedicated link connects two devices, reserving full capacity for their communication. Examples include microwave or satellite links, or an infrared remote control connecting to a TV.
 - **Multipoint:** Multiple devices share a single link, either **spatially** (simultaneous use, like Wi-Fi) or **temporally** (time-shared, like a bus network). This reduces cabling but requires coordination to avoid conflicts.

Figure 1.3 Types of connections: point-to-point and multipoint



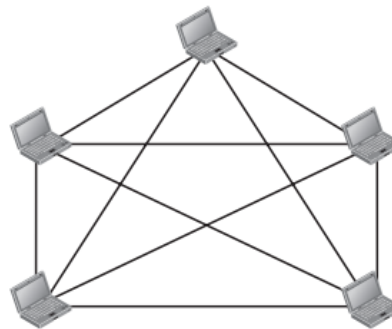
- **Physical Topology** (Figure 1.4–1.7): Topologies define the physical or logical arrangement of devices:

1. **Mesh Topology:**

- Each device has a dedicated point-to-point link to every other device, requiring $(n(n-1)/2)$ links for (n) nodes in duplex mode.
- **Advantages:** Eliminates traffic congestion (dedicated links), robust (one link failure doesn't affect others), secure (private links), and easy to isolate faults.
- **Disadvantages:** Requires extensive cabling and I/O ports, making installation/reconfiguration complex and costly. Space constraints and hardware expenses limit its use.
- **Example:** Interconnections between telephone regional offices, where high reliability and security are critical.

Figure 1.4 A fully connected mesh topology (five devices)

$n = 5$
10 links.

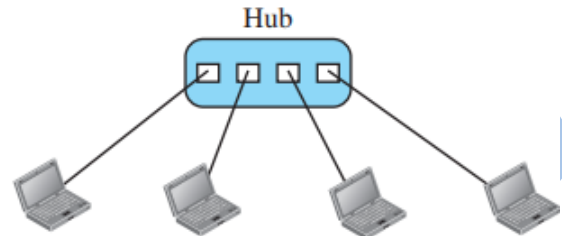


2. **Star Topology:**

- Devices connect to a central hub via point-to-point links, with the hub relaying data.
- **Advantages:** Requires less cabling than mesh, easy to install/reconfigure, robust (single link failure affects only that device), and simple fault isolation via the hub.

- **Disadvantages:** Hub failure disables the entire network, and cabling to the hub may still be significant compared to other topologies.
- **Example:** Used in high-speed LANs, such as Ethernet networks with a central switch.

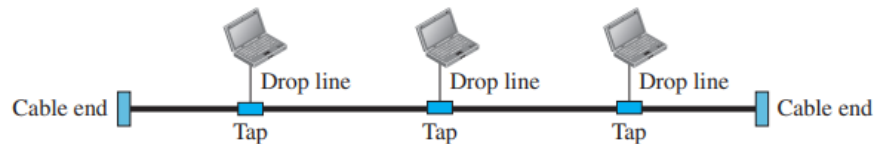
Figure 1.5 *A star topology connecting four stations*



3. Bus Topology:

- A single backbone cable connects all devices via drop lines and taps.
- **Advantages:** Easy to install, minimal cabling (only the backbone and drop lines).
- **Disadvantages:** Difficult to reconfigure or isolate faults, signal degradation due to energy loss, and a backbone failure halts all communication. Limited tap numbers and spacing.
- **Example:** Early Ethernet LANs, now less common due to these limitations.

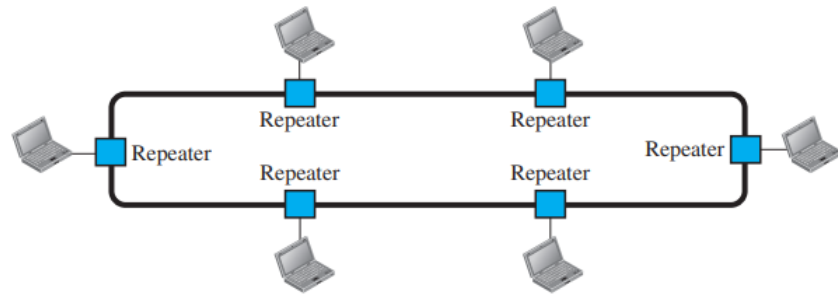
Figure 1.6 *A bus topology connecting three stations*



4. Ring Topology:

- Each device connects to two neighbors, forming a ring where signals pass unidirectionally via repeaters.
- **Advantages:** Easy to install/reconfigure (only two connections per device), simplified fault isolation (devices detect signal absence).
- **Disadvantages:** Unidirectional traffic can be limiting, and a single break can disable the network unless mitigated by dual rings or switches.
- **Example:** IBM's Token Ring LAN, though less popular today due to speed limitations.

Figure 1.7 A ring topology connecting six stations

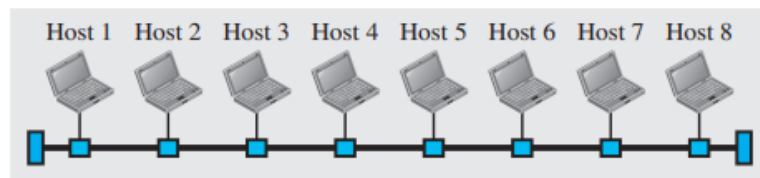


Network Types:

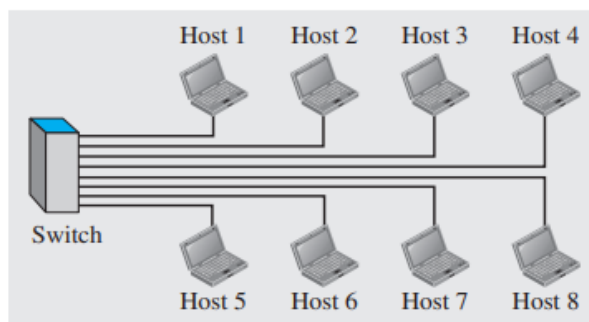
1. Local Area Network (LAN):

- Privately owned, spans a small area (e.g., office, building, campus).
- Historically used a common cable, but modern LANs use switches to direct packets efficiently, reducing congestion and enabling simultaneous communication.
- Hosts have unique addresses, and switches guide packets to their destinations.
- **Example:** A company's office network sharing printers and files.

Figure 1.8 An isolated LAN in the past and today

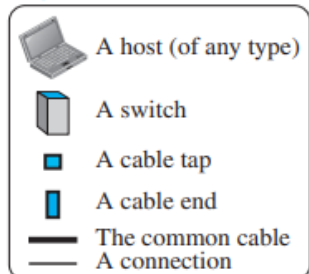


a. LAN with a common cable (past)



b. LAN with a switch (today)

Legend



2. Wide Area Network (WAN):

- Spans large geographical areas (e.g., city, country, world), connecting devices like routers and switches.
- Often leased from communication providers, unlike privately owned LANs.
- **Types:**

- **Point-to-Point WAN:** Connects two devices (e.g., a leased line between offices).
 - **Switched WAN:** Connects multiple devices via switches, forming the backbone of global communication.
- **Example:** Internet Service Provider (ISP) networks connecting cities.

Figure 1.9 A point-to-point WAN

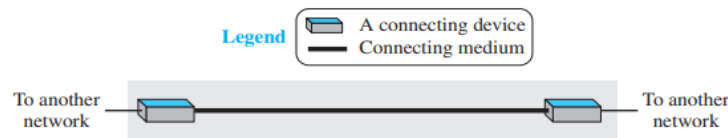
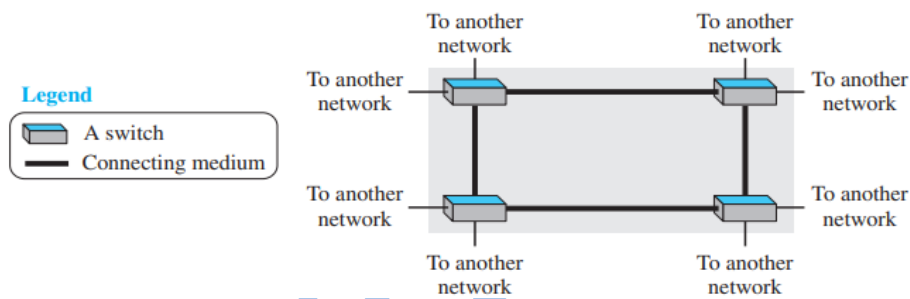


Figure 1.10 A switched WAN



3. Internetwork (Internet):

- A network of networks, connecting LANs and WANs (e.g., two office LANs linked via a point-to-point WAN).
- Routers route packets between networks; switches handle intra-LAN traffic.
- **Example:** A company's private internet connecting east and west coast offices.

Figure 1.11 An internetwork made of two LANs and one point-to-point WAN

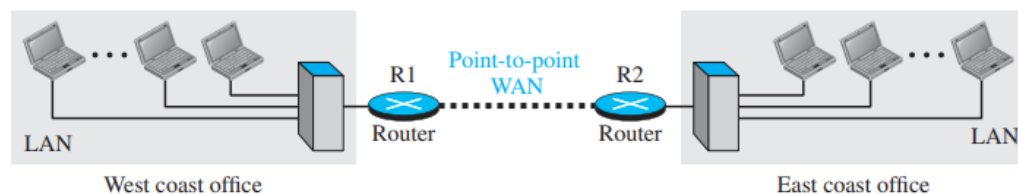
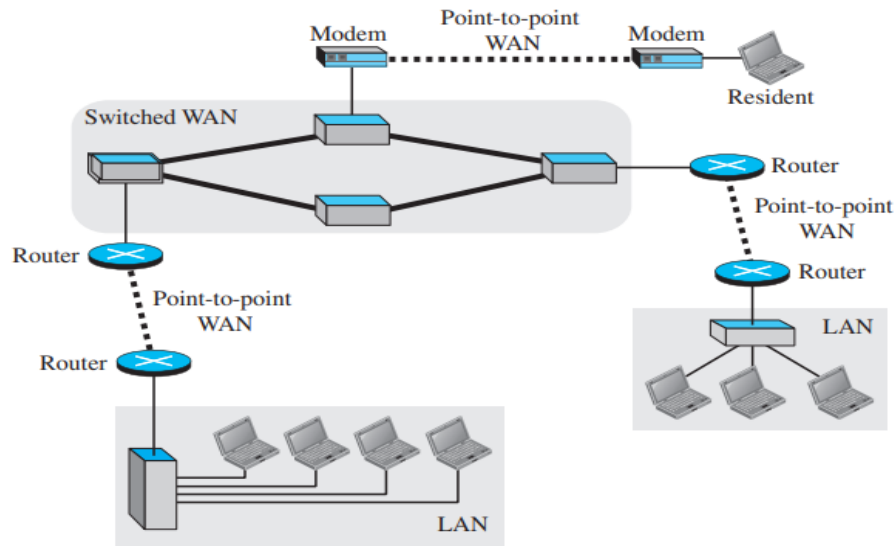


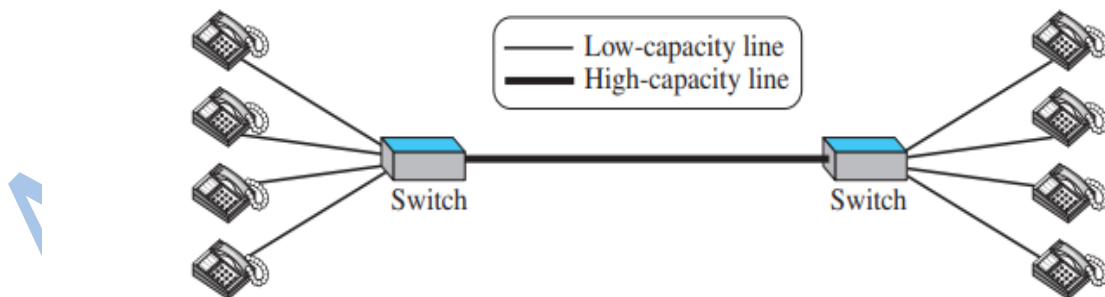
Figure 1.12 *A heterogeneous network made of four WANs and three LANs*



Switching:

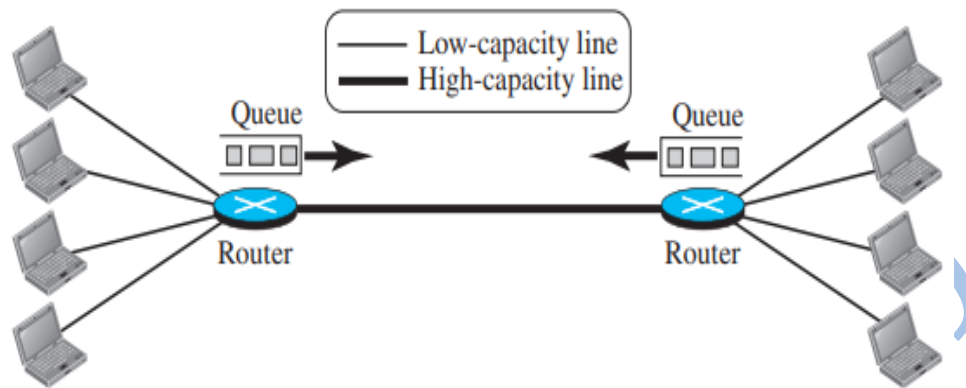
- **Circuit-Switched Network:** Establishes a dedicated circuit between endpoints, ideal for constant-rate communication like traditional telephony. Inefficient at partial capacity, as unused bandwidth is wasted.

Figure 1.13 *A circuit-switched network*



- **Packet-Switched Network:** Divides data into packets, which are stored and forwarded by routers. More efficient, as bandwidth is shared dynamically, but packets may face delays due to queuing.

Figure 1.14 *A packet-switched network*



The Internet:

- A global internetwork of backbones (large ISP networks like AT&T), provider networks (smaller ISPs), and customer networks (end users).
- Backbones connect via peering points; provider/customer networks pay for access.
- **Access Methods:**
 - **Telephone Networks:**
 - **Dial-Up:** Converts data to voice via a modem, slow and blocks voice calls.
 - **DSL:** Uses existing phone lines for high-speed data, allowing simultaneous voice use.
 - **Cable Networks:** High-speed but shared among users, causing variable performance.
 - **Wireless Networks:** Combine wired/wireless or use wireless WANs for flexible connectivity.
 - **Direct Connection:** Large organizations lease high-speed WANs to connect directly to ISPs.

Figure 1.15 *The Internet today*

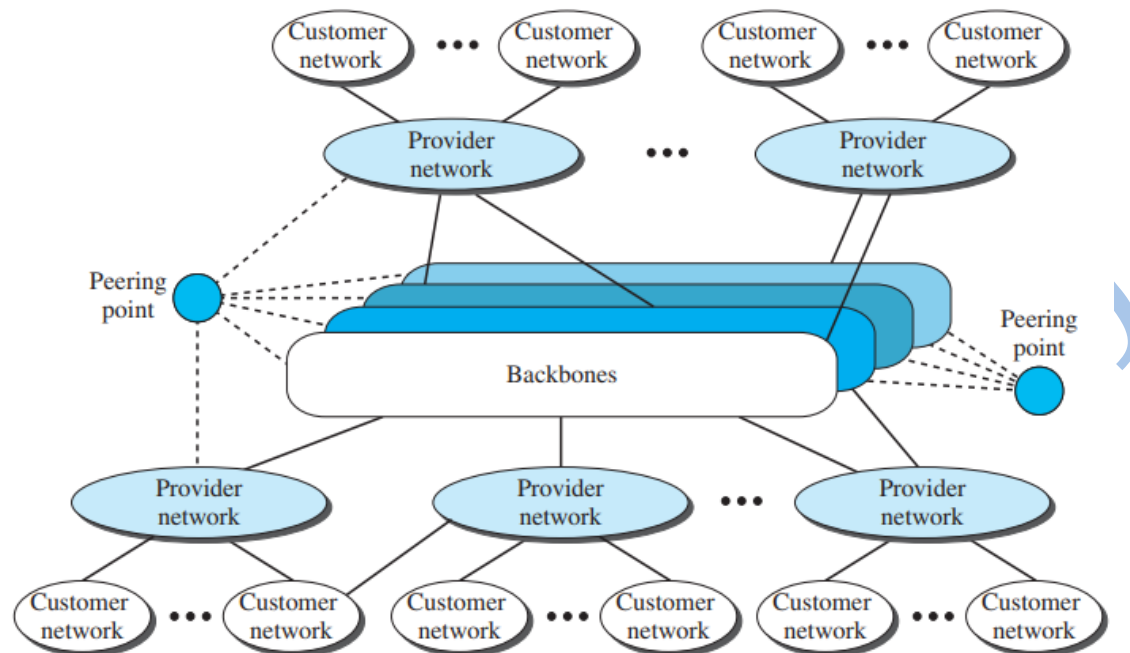


Figure 1.15 shows a conceptual (not geographical) view of the Internet.

The figure shows the Internet as several backbones, provider networks, and customer networks. At the top level, the backbones are large networks owned by some communication companies such as Sprint, Verizon (MCI), AT&T, and NTT. The backbone networks are connected through some complex switching systems, called peering points. At the second level, there are smaller networks, called provider networks, that use the services of the backbones for a fee.

The provider networks are connected to backbones and sometimes to other provider networks. The customer networks are networks at the edge of the Internet that actually use the services provided by the Internet. They pay fees to provider networks for receiving services.

Backbones and provider networks are also called Internet Service Providers (ISPs). The backbones are often referred to as international ISPs; the provider networks are often referred to as national or regional ISPs.

Internet History:

- Pre-1960 networks (telegraph, telephone) supported constant-rate communication, unsuitable for bursty data.
- Packet-switched networks, developed post-1960, enabled the Internet's growth into a global, scalable system.

Additional Explanation: Networks are the infrastructure enabling modern connectivity, from small-scale LANs to the global Internet. The criteria of performance, reliability, and security guide network design to meet user needs. Topologies like mesh offer high reliability but are costly, while star and bus topologies balance cost and efficiency. The shift from circuit-switched to packet-switched networks reflects the need for efficient, flexible data transfer in modern applications. Understanding network types and access methods is crucial for appreciating how the Internet connects billions of devices, supporting everything from email to streaming services.

3. Protocol Layering

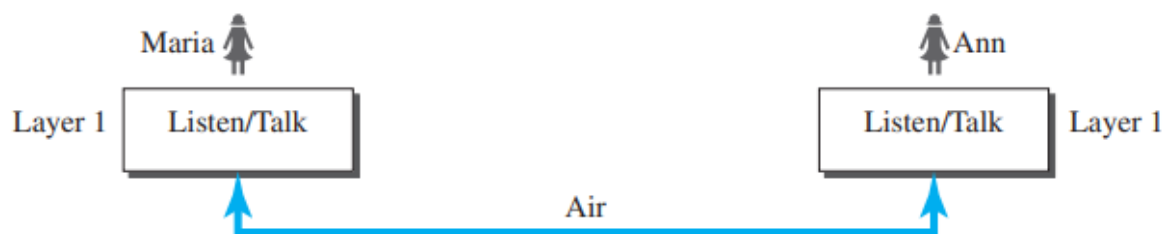
Definition: A protocol is a set of rules governing data communication. Protocol layering divides complex communication tasks into simpler, modular layers, each with a specific protocol. This approach enhances manageability, scalability, and flexibility in network design.

Scenarios:

1. **Single-Layer Scenario** (Figure 2.1):

- Simple communication, like a face-to-face conversation, uses one layer with basic rules (e.g., greeting, turn-taking, vocabulary limits).
- **Example:** Two people discussing ideas follow implicit protocols to ensure mutual understanding, such as pausing to listen or maintaining polite dialogue.

Figure 2.1 *A single-layer protocol*

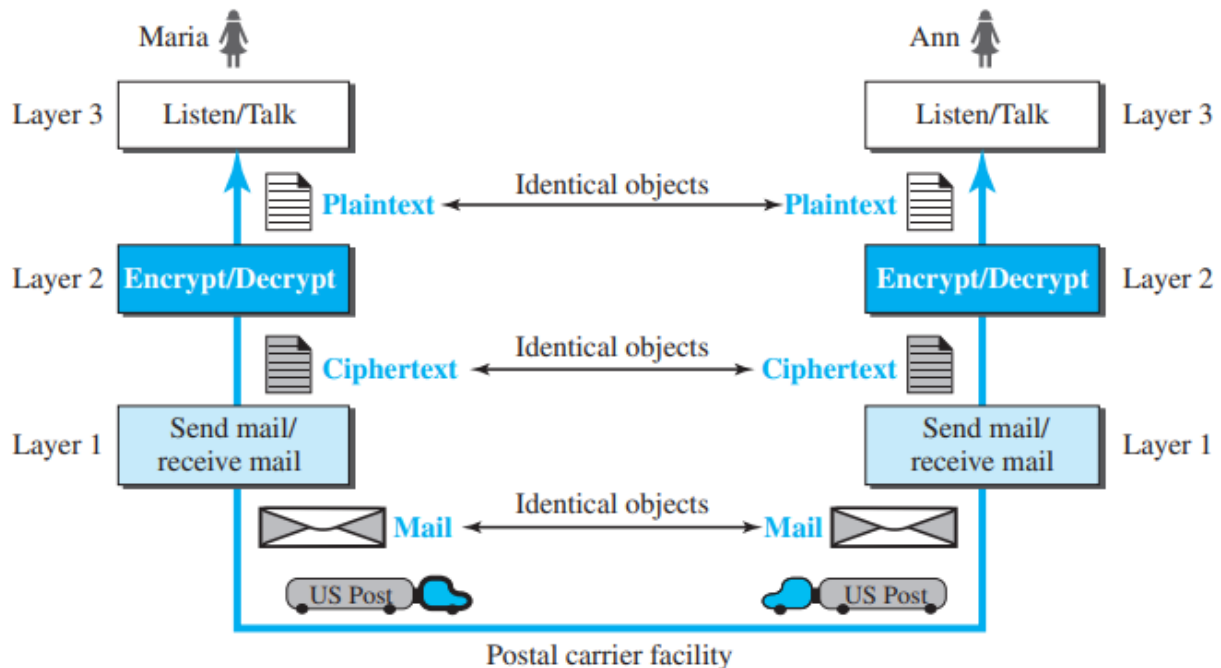


2. **Multi-Layer Scenario** (Figure 2.2):

- Complex communication, like encrypted mail, requires multiple layers:
 - **Layer 3 (Application):** Creates plaintext (e.g., a letter's content).
 - **Layer 2 (Encryption):** Encrypts plaintext to ciphertext for security.
 - **Layer 1 (Physical Delivery):** Sends/receives the physical mail.
- **Advantages:**
 - **Modularity:** Each layer is independent, allowing updates (e.g., changing encryption) without affecting others.
 - **Service Separation:** Layers focus on specific tasks, abstracting implementation details. For example, the application layer doesn't need to know how encryption works.

- **Intermediate Systems:** Devices like routers only need lower layers, reducing complexity and cost.

Figure 2.2 *A three-layer protocol*



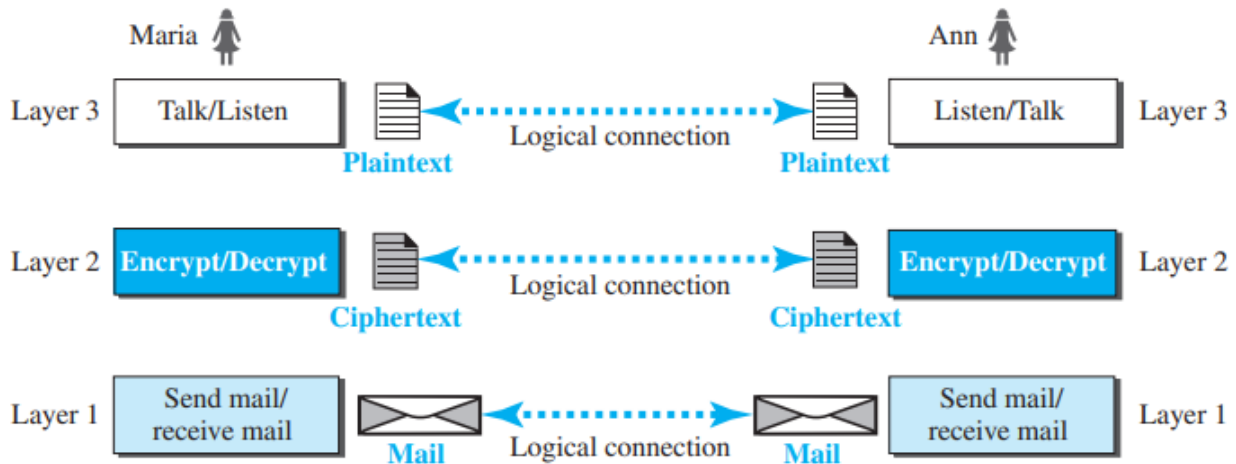
Principles of Protocol Layering:

1. **Bidirectional Tasks:** Each layer performs two opposite tasks (e.g., sending/receiving, encrypting/decrypting) to support two-way communication.
2. **Identical Objects:** Objects at each layer (e.g., plaintext, ciphertext) are identical at both sender and receiver, ensuring compatibility.

Logical Connections (Figure 2.3):

- Layers appear to communicate directly (logically) with their counterparts at the other end, though data physically passes through all layers. This abstraction simplifies understanding and design, as each layer focuses on its specific role.

Figure 2.3 Logical connection between peer layers



Additional Explanation: Protocol layering is a cornerstone of modern networking, allowing complex systems to be broken into manageable parts. The single-layer scenario illustrates basic communication, while the multi-layer scenario highlights how layering handles complexity, such as securing data over long distances. Modularity ensures that advancements in one layer (e.g., new encryption algorithms) don't disrupt others, promoting innovation. Logical connections abstract the physical complexity, making it easier to design and troubleshoot networks, as seen in the Internet's layered architecture.

4. TCP/IP Protocol Suite

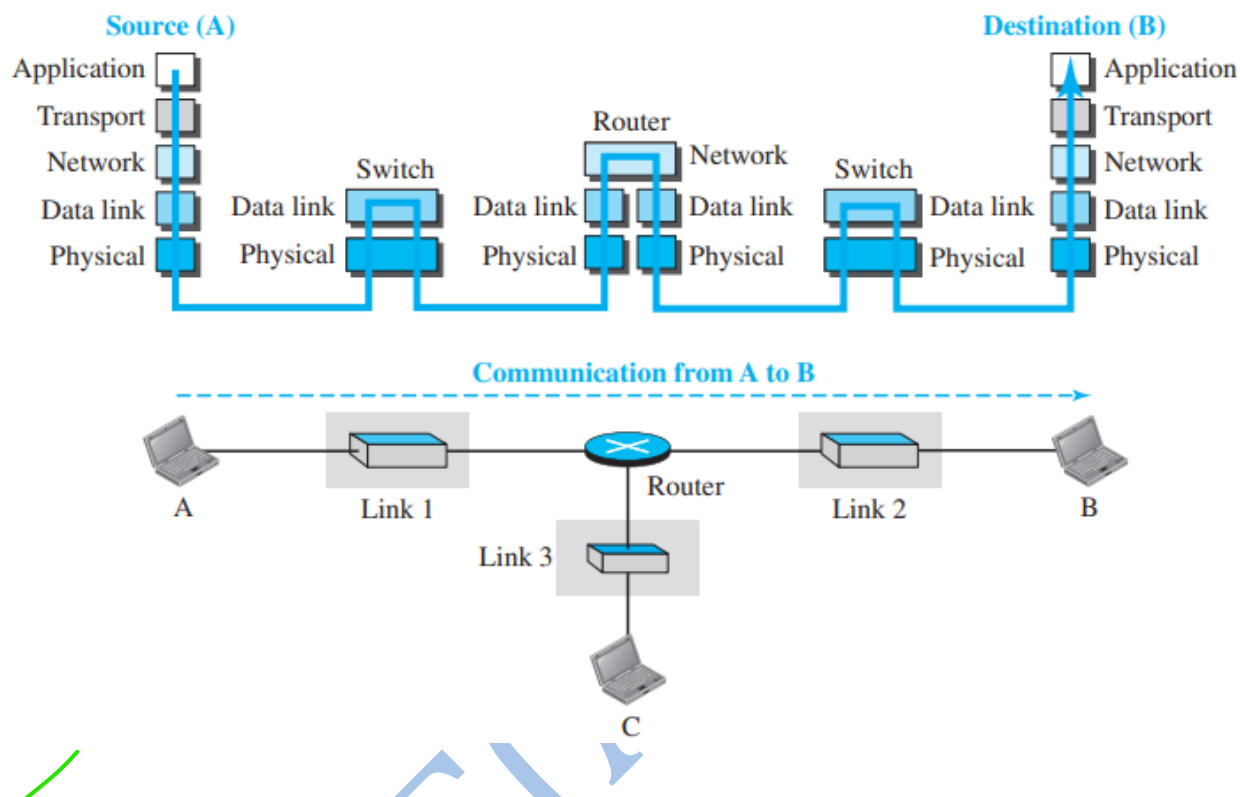
Definition: The TCP/IP (Transmission Control Protocol/Internet Protocol) suite is a hierarchical set of protocols powering the Internet. It organizes communication into five layers: Physical, Data-Link, Network, Transport, and Application, each providing specific functions.

Layered Architecture (Figure 2.4, 2.5):

- **Devices:** Source/destination hosts use all five layers; routers use physical, data-link, and network layers; link-layer switches use physical and data-link layers.
- **Logical Connections** (Figure 2.6): Application, transport, and network layers provide end-to-end communication (source to destination); data-link and physical layers are hop-to-hop (between adjacent devices).

X

Figure 2.5 *Communication through an internet*



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Figure 2.6 *Logical connections between layers of the TCP/IP protocol suite*

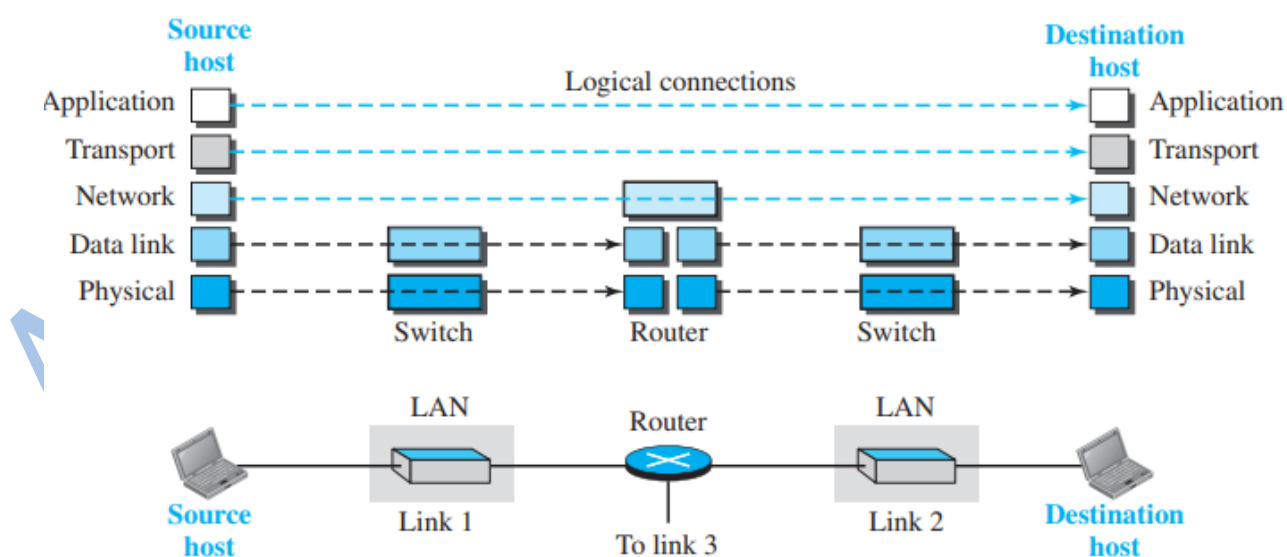


Figure 2.7 *Identical objects in the TCP/IP protocol suite*

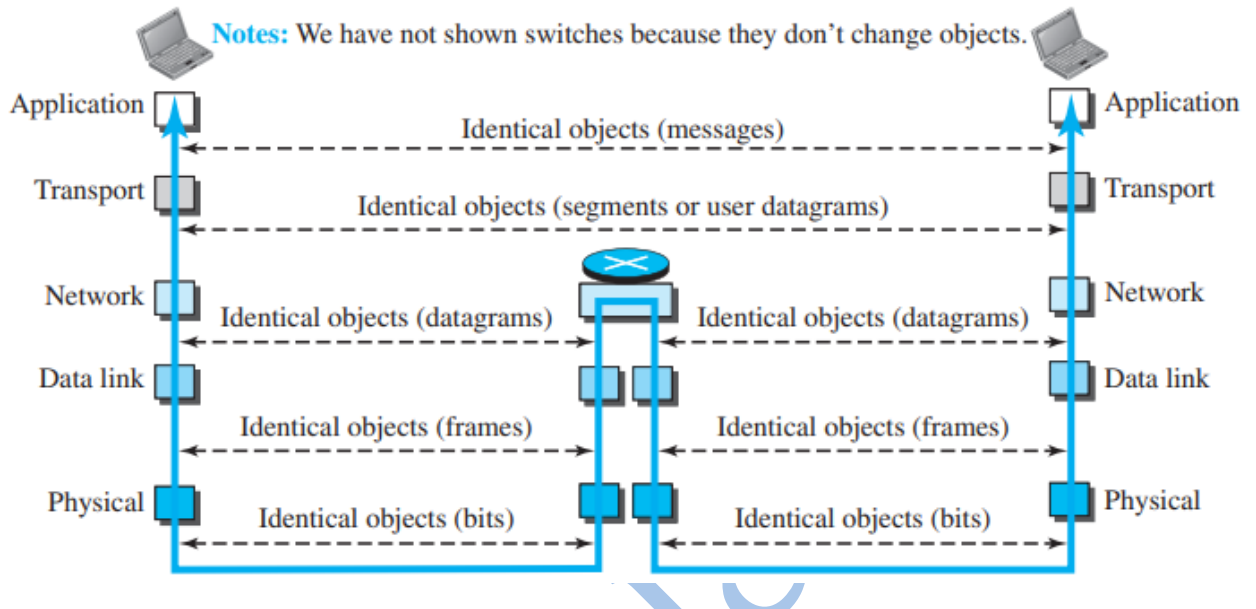
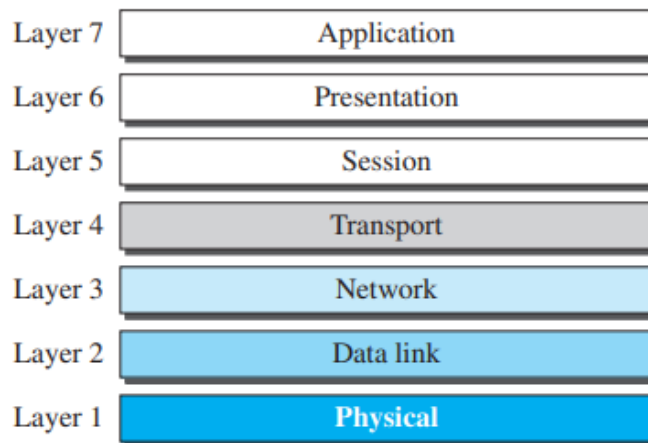


Figure 2.11 *The OSI model*



Layers and Functions:

Application Layer

- **Responsibility:** Serves as the interface between **user** applications and the **network**, handling high-level **protocols** and data **formatting** for user-facing services.
- **Functions:** Enables applications like web browsers, email clients, and file transfer programs to communicate over the network. It manages data exchange, presentation, and session management (e.g., initiating and terminating connections). Protocols include **HTTP** (web browsing), **SMTP** (email), **FTP** (file transfer), and DNS (domain name resolution).

- **Example:** When you access a website, the application layer formats your HTTP request and processes the server's response for display in your browser.

Transport Layer

- **Responsibility:** Ensures **end-to-end** communication, managing data **transfer** reliability, flow control, and **error correction** between devices.
- **Functions:** Segments data into **packets** and assigns **port numbers** to identify applications. Uses **TCP** for **reliable**, connection-oriented communication with retransmission and error checking, or **UDP** for **faster**, connectionless communication. It ensures data is delivered in the correct order and without loss (for TCP).
- **Example:** TCP ensures all parts of a downloaded file arrive correctly, while UDP supports real-time video streaming where speed trumps occasional packet loss.

Network Layer

- **Responsibility:** Handles **logical** addressing and **routing** of data packets across different networks to reach their destination.
- **Functions:** Uses **IP** (IPv4 or IPv6) to assign **source** and **destination** addresses and **route** packets through networks via routers. Supports protocols like **ICMP** (error messaging) and **ARP** (address resolution). It determines the best path for data delivery.
- **Example:** When sending an email, the network layer routes packets from your device's IP address to the recipient's IP address across the internet.

Data Link Layer

- **Responsibility:** Manages reliable data **transfer** across a single physical link, handling **framing**, **error** detection, and physical **addressing**.
- **Functions:** Formats data into **frames** and uses MAC addresses to identify devices on the same network. Protocols like **Ethernet and Wi-Fi** operate here, handling error detection (e.g., CRC) and access control to **shared media** (e.g., CSMA/CD for Ethernet).
- **Example:** In a local Wi-Fi network, the data link layer ensures data frames are sent to the correct device (e.g., your laptop's MAC address) without collisions.

Physical Layer

- **Responsibility:** Governs the physical **transmission of raw bits** over a medium, such as cables, fiber optics, or wireless signals.
- **Functions:** Defines **hardware specifications** like voltage levels, cable types, and signal modulation. It **transmits bits** (0s and 1s) without interpreting their meaning, handling connectors, bit rates, and physical topologies (e.g., star, bus).

- **Example:** When data travels over an Ethernet cable, the physical layer converts frames into electrical signals for transmission to the router.

Detailed Explanation (Write based on marks)

1. Application Layer

Responsibility

The **Application Layer** is the topmost layer in the OSI model, serving as the interface between user applications and the network. It enables software applications (e.g., web browsers, email clients) to communicate by providing protocols and services for data exchange, ensuring data is formatted and presented in a way that applications and users can understand.

Functions

- **Protocol Support:** Manages application-specific protocols, including:
 - **HTTP/HTTPS:** For web browsing (e.g., accessing websites).
 - **SMTP/POP3/IMAP:** For sending (SMTP) and receiving (POP3/IMAP) emails.
 - **FTP/SFTP:** For file transfers, with SFTP adding security via SSH.
 - **DNS:** Resolves domain names (e.g., x.ai) to IP addresses.
 - **Telnet/SSH:** For remote terminal access, with SSH providing encryption.
 - **SNMP:** For network device management.
- **Data Formatting and Presentation:** Ensures data is in a usable format (e.g., text as ASCII, images as JPEG). This overlaps with OSI's Presentation Layer, handling tasks like encryption (e.g., TLS for HTTPS), compression, and character encoding.
- **Session Management:** Initiates, maintains, and terminates communication sessions (overlapping with OSI's Session Layer). For example, it manages cookies for web sessions or authentication for email access.
- **User Interaction:** Provides the interface for applications to send/receive data, abstracting network complexities.

Process

The Application Layer generates data based on user actions (e.g., typing a URL). It formats the data, adds protocol-specific headers (e.g., HTTP headers), and passes it to the Transport Layer. On the receiving end, it interprets incoming data and delivers it to the appropriate application (e.g., rendering a webpage in a browser).

Example

- **Scenario:** You visit <https://x.ai> in a browser.
 - The browser (via HTTP/HTTPS) creates a request ("GET /index.html").

- The Application Layer formats the request, encrypts it (TLS for HTTPS), and sends it to the Transport Layer.
 - The server's response (HTML data) is received, decrypted, and rendered as a webpage.
- **DNS Example:** The Application Layer uses DNS to resolve "x.ai" to an IP address (e.g., 172.16.254.1) before sending the request.

Significance

The Application Layer is the user's gateway to the network, supporting services like web browsing, emailing, and file sharing. In TCP/IP, it encompasses OSI's Application, Presentation, and Session layers, making it broader and more practical for real-world implementations.

2. Transport Layer

Responsibility

The **Transport Layer** ensures **end-to-end communication** between devices, managing data transfer reliability, segmentation, flow control, and error handling. It acts as a mediator between the Application Layer and lower layers, ensuring data reaches the correct application.

Functions

- **Segmentation and Reassembly:** Breaks application data into smaller segments for transmission and reassembles them at the destination.
- **Protocol Options:**
 - **TCP (Transmission Control Protocol):**
 - **Connection-Oriented:** Establishes a connection using a three-way handshake (SYN, SYN-ACK, ACK).
 - **Reliability:** Ensures data delivery without loss or duplication via sequence numbers, acknowledgments, and retransmissions.
 - **Flow Control:** Uses sliding windows to prevent overwhelming the receiver.
 - **Error Checking:** Detects and corrects errors using checksums.
 - **UDP (User Datagram Protocol):**
 - **Connectionless:** Sends datagrams without establishing a connection, prioritizing speed over reliability.
 - **Low Overhead:** Suitable for real-time applications (e.g., video streaming, gaming) where occasional packet loss is acceptable.
- **Port Numbers:** Assigns source and destination port numbers to identify applications (e.g., port 80 for HTTP, 443 for HTTPS, 53 for DNS).

- **Multiplexing/Demultiplexing:** Enables multiple applications to share the network by directing data to the correct application based on ports.
- **Congestion Control:** TCP adjusts transmission rates to avoid network congestion.

Process

The Transport Layer receives data from the Application Layer, segments it, adds a header (with port numbers, sequence numbers for TCP, or checksums), and passes it to the Network Layer. On the receiving end, it reassembles segments, verifies integrity, and forwards data to the correct application via ports.

Example

- **TCP Scenario:** Downloading a file (e.g., a 10 MB PDF).
 - The Transport Layer segments the file into packets (e.g., 1500-byte segments), adds TCP headers with sequence numbers, and ensures all packets arrive correctly by retransmitting lost ones.
 - The receiver acknowledges each packet, and TCP reassembles the file.
- **UDP Scenario:** Streaming a live video on a platform like Twitch.
 - UDP sends video data as datagrams without retransmission, ensuring low latency. Minor packet loss may cause brief glitches but maintains real-time playback.

Significance

The Transport Layer provides flexibility by offering reliable (TCP) and lightweight (UDP) options, catering to diverse application needs. It ensures data integrity and application-specific delivery, shielding applications from network-level issues. In TCP/IP, this layer is directly equivalent to OSI's Transport Layer.

3. Network Layer

Responsibility

The **Network Layer** handles **logical addressing** and **routing**, ensuring data packets are delivered from the source to the destination across multiple networks. It enables communication between devices on different networks, forming the backbone of internet connectivity.

Functions

- **IP Addressing:** Assigns unique addresses to devices:
 - **IPv4:** 32-bit addresses (e.g., 192.168.1.1), supporting ~4.3 billion devices.

- **IPv6:** 128-bit addresses (e.g., 2001:0db8::1), supporting vastly more devices to accommodate internet growth.
- **Packet Routing:** Determines the optimal path for packets using routing protocols (e.g., BGP, OSPF) and routers. Routers use routing tables to forward packets across networks.
- **Packet Encapsulation:** Adds IP headers (with source and destination IP addresses) to Transport Layer segments, forming packets (or datagrams for UDP).
- **Fragmentation and Reassembly:** Breaks large packets into smaller fragments to fit network-specific MTU (Maximum Transmission Unit) sizes and reassembles them at the destination.
- **Supporting Protocols:**
 - **ICMP:** Handles error messages and diagnostics (e.g., ping uses ICMP to test connectivity).
 - **ARP:** Maps IP addresses to MAC addresses within a local network.
 - **IGMP:** Manages multicast group membership for efficient data delivery.

Process

The Network Layer receives segments from the Transport Layer, encapsulates them into packets with IP headers, and forwards them to the Data Link Layer. Routers use IP addresses to route packets across networks (e.g., from your ISP to a remote server). On the receiving end, the Network Layer strips the IP header and passes the segment to the Transport Layer.

Example

- **Scenario:** Sending an email to a friend in another country.
 - The Network Layer assigns your device's IP (e.g., 192.168.1.100) as the source and the email server's IP (e.g., 172.16.254.1) as the destination.
 - Routers forward packets through multiple networks, choosing the best path based on routing tables.
 - If a packet exceeds the MTU (e.g., 1500 bytes), the Network Layer fragments it, and the destination reassembles it.

Significance

The Network Layer enables global connectivity by standardizing addressing and routing. In TCP/IP, it corresponds to the **Internet Layer**, which is critical for the internet's operation, allowing devices on different networks to communicate seamlessly.

4. Data Link Layer

Responsibility

The **Data Link Layer** manages **reliable data transfer** across a single physical link (e.g., between two devices on the same network). It handles framing, physical addressing, error detection, and access control to shared media.

Functions

- **Framing:** Organizes bits from the Physical Layer into frames, adding headers (e.g., source and destination MAC addresses) and trailers (e.g., CRC for error detection).
- **Physical Addressing:** Uses **MAC addresses** (48-bit unique identifiers, e.g., 00:1A:2B:3C:4D:5E) to identify devices on the same network.
- **Error Detection:** Includes mechanisms like **CRC (Cyclic Redundancy Check)** to detect transmission errors caused by noise or interference. Errors are typically reported to higher layers for correction.
- **Media Access Control (MAC):** Manages how devices share the physical medium to avoid collisions:
 - **CSMA/CD** (Carrier Sense Multiple Access/Collision Detection): Used in Ethernet to detect and resolve collisions.
 - **CSMA/CA** (Collision Avoidance): Used in Wi-Fi to prevent collisions.
- **Flow Control:** Prevents overwhelming the receiver (e.g., using pause frames in Ethernet).
- **Protocols and Standards:** Includes Ethernet (802.3), Wi-Fi (802.11), PPP (Point-to-Point Protocol), and HDLC.

Process

The Data Link Layer receives packets from the Network Layer, encapsulates them into frames with MAC addresses, and passes them to the Physical Layer for transmission. It ensures frames are transmitted error-free over the physical link. On the receiving end, it checks frames for errors, strips headers/trailers, and passes packets to the Network Layer.

Example

- **Scenario:** Sending data over a local Ethernet network.
 - The Data Link Layer encapsulates an IP packet into an Ethernet frame, adding your device's MAC address (source) and the router's MAC address (destination).
 - CRC is added to detect errors. CSMA/CD ensures no collisions occur on the shared medium.
 - The frame is sent to the Physical Layer for bit-level transmission.

- **Wi-Fi Example:** In a Wi-Fi network, CSMA/CA prevents collisions, and frames are sent to the access point's MAC address.

Significance

The Data Link Layer ensures reliable communication within a local network, bridging logical packets to physical transmission. In TCP/IP, it is part of the **Network Access Layer**, combining Data Link and Physical Layer functions.

5. Physical Layer

Responsibility

The **Physical Layer** governs the **physical transmission of raw bits** (0s and 1s) over a physical medium, such as copper cables, fiber optics, or wireless signals. It handles the hardware and electrical/optical properties required to send and receive bits without interpreting their meaning.

Functions

- **Bit Transmission:** Converts frames from the Data Link Layer into raw bits and transmits them as physical signals:
 - **Electrical Signals:** In copper cables (e.g., Ethernet), bits are represented as voltage levels (e.g., 0V for 0, 2.5V for 1).
 - **Optical Signals:** In fiber optics, bits are light pulses (e.g., no light for 0, light for 1).
 - **Radio Signals:** In wireless (e.g., Wi-Fi), bits are modulated onto radio waves using techniques like QAM or PSK.
- **Signal Encoding:** Uses encoding schemes to represent bits reliably:
 - **Manchester Encoding:** Transitions (high-to-low or low-to-high) represent 0s and 1s, aiding synchronization.
 - **NRZ (Non-Return-to-Zero):** Direct voltage levels for 0s and 1s.
 - **QAM (Quadrature Amplitude Modulation):** Encodes multiple bits per symbol in wireless networks (e.g., 256-QAM encodes 8 bits per symbol).
- **Hardware Specifications:**
 - **Media:** Defines cables (e.g., Cat6, fiber optic) or wireless bands (e.g., 2.4 GHz, 5 GHz for Wi-Fi).
 - **Connectors:** Specifies interfaces like RJ45 (Ethernet), SC/LC (fiber), or antennas.
 - **Bit Rate:** Determines transmission speed (e.g., 1 Gbps for Gigabit Ethernet, 10 Gbps for fiber).
- **Synchronization:** Ensures sender and receiver align on bit timing using clock signals or encoding transitions.

- **Physical Topology:** Defines network layouts (e.g., star, bus, ring).
- **Signal Integrity:** Manages issues like attenuation (signal weakening), noise, and interference using amplifiers, repeaters, or shielded cables.

Process

The Physical Layer receives frames from the Data Link Layer, converts them into a stream of bits, and transmits them as signals over the medium. On the receiving end, it converts signals back into bits, reconstructs frames, and passes them to the Data Link Layer.

Example

- **Scenario:** Transmitting data over an Ethernet cable.
 - The Data Link Layer provides a frame containing the bits for “A” (ASCII 65, binary 01000001).
 - The Physical Layer converts these bits into electrical signals (e.g., low voltage for 0, high voltage for 1) and sends them over a Cat6 cable.
 - The receiver’s Physical Layer decodes the voltages back into bits, passing the frame to the Data Link Layer.
- **Wi-Fi Example:** Bits are modulated onto a 5 GHz signal using 256-QAM, where each symbol represents 8 bits, enabling high-speed Wi-Fi (e.g., 802.11ac).
- **Bit-Level Detail:** For a 1 MB file (8 million bits), a 1 Gbps Ethernet link transmits it in ~8 ms (excluding overhead). Noise or interference could flip a bit (e.g., 0 to 1), but the Data Link Layer’s CRC detects this.

Significance

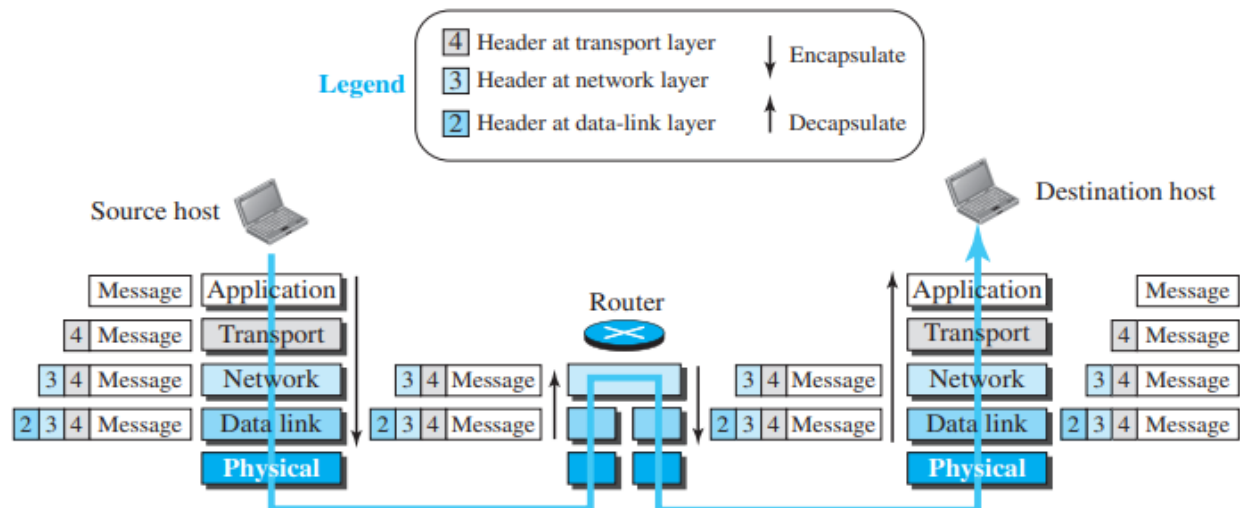
The Physical Layer is the foundation of networking, handling the raw transmission of bits. In TCP/IP, it’s part of the **Network Access Layer**, working with the Data Link Layer to connect logical data to physical media. Its role in encoding and transmitting bits ensures all higher-layer functions can operate.

Mapping to TCP/IP Model

- **Application Layer (OSI):** Maps to TCP/IP’s **Application Layer**, which also includes OSI’s Presentation and Session layers (e.g., encryption, session management).
- **Transport Layer (OSI):** Directly corresponds to TCP/IP’s **Transport Layer** (TCP/UDP).
- **Network Layer (OSI):** Maps to TCP/IP’s **Internet Layer** (IP, ICMP, ARP).
- **Data Link and Physical Layers (OSI):** Combined into TCP/IP’s **Network Access Layer**, covering framing, MAC addressing, and bit transmission.

Encapsulation/Decapsulation (Figure 2.8):

Figure 2.8 *Encapsulation/Decapsulation*



- **Source Host:**
 - **Application:** Creates a message (e.g., email content).
 - **Transport:** Adds a header (e.g., port numbers), forming a segment/user datagram.
 - **Network:** Adds a header (e.g., IP addresses), forming a datagram.
 - **Data-Link:** Adds a header (e.g., MAC addresses), forming a frame.
 - **Physical:** Transmits bits as signals.
- **Router:** Decapsulates frame to datagram, routes based on IP address, re-encapsulates for the next link.
- **Destination Host:** Decapsulates frame through layers, delivering the message to the application, with error checking at each step.

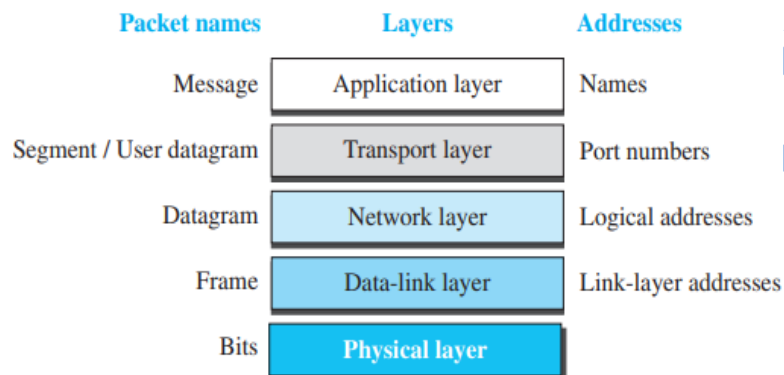
Encapsulation Example

- **Data:** The string “Hello” (5 bytes).
- **Application Layer:** Adds HTTP header (e.g., “GET /hello”).
- **Transport Layer:** Adds TCP header (e.g., source port 12345, destination port 80).
- **Network Layer:** Adds IP header (e.g., source IP 192.168.1.100, destination IP 172.16.254.1).
- **Data Link Layer:** Adds Ethernet frame header (MAC addresses) and trailer (CRC).
- **Physical Layer:** Transmits bits as electrical signals (e.g., 01001000 for “H”).

Addressing (Figure 2.9):

- **Application:** Uses names (e.g., someorg.com) for user-friendly identification.
- **Transport:** Uses port numbers to identify specific processes (e.g., port 80 for HTTP).
- **Network:** Uses logical (IP) addresses for global host identification.
- **Data-Link:** Uses MAC addresses for local network identification.

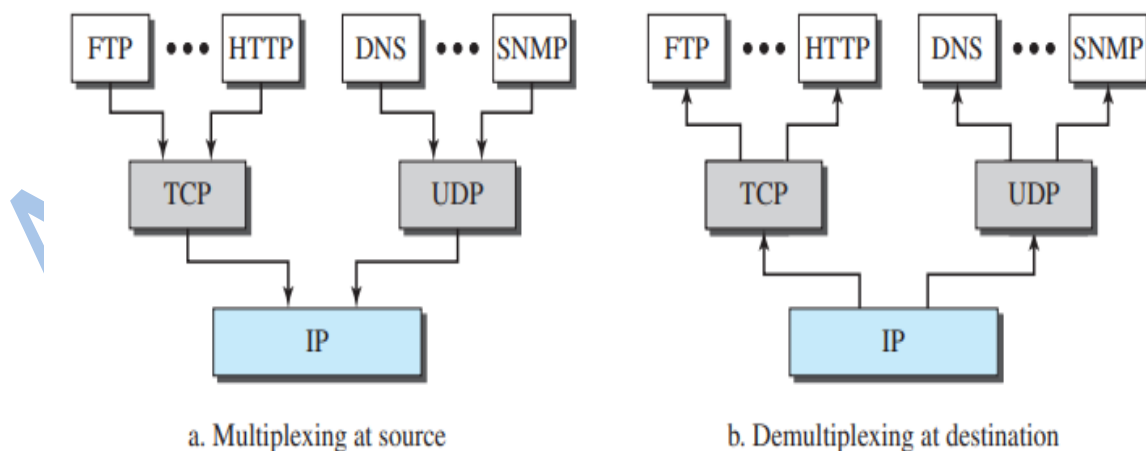
Figure 2.9 Addressing in the TCP/IP protocol suite



Multiplexing/Demultiplexing (Figure 2.10):

- **Multiplexing:** A layer encapsulates packets from multiple higher-layer protocols (e.g., IP handles TCP, UDP, ICMP).
- **Demultiplexing:** A layer delivers packets to the correct higher-layer protocol using header fields.

Figure 2.10 Multiplexing and demultiplexing



Additional Explanation: The TCP/IP suite is the foundation of the Internet, designed for flexibility and scalability. Each layer focuses on a specific task, from bit transmission to application communication, ensuring modularity and interoperability. Encapsulation/decapsulation allows data to traverse diverse networks seamlessly, while

addressing ensures precise delivery. Multiplexing/demultiplexing enables efficient use of shared resources, supporting multiple applications simultaneously. The suite's hierarchical nature allows devices like routers to focus on lower layers, optimizing performance and cost.

5. OSI Model

Definition: The Open Systems Interconnection (OSI) model, developed by the International Organization for Standardization (ISO), is a **seven**-layer framework for designing interoperable network systems. It is a conceptual model, not a protocol, aimed at **standardizing communication** across diverse systems.

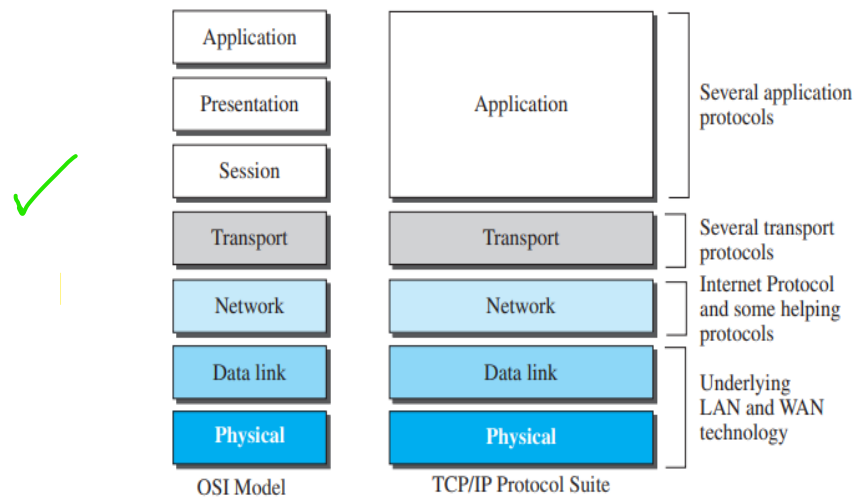
Layers (Figure 2.11):

1. **Physical:** Transmits raw bits over a medium.
2. **Data-Link:** Ensures error-free frame delivery across a link.
3. **Network:** Routes packets from source to destination.
4. **Transport:** Provides reliable process-to-process delivery.
5. **Session:** Manages sessions (e.g., initiating, maintaining, terminating connections).
6. **Presentation:** Handles data formatting, encryption, and compression.
7. **Application:** Provides network services to user applications.

OSI vs. TCP/IP (Figure 2.12):

- TCP/IP combines OSI's session and presentation layers into its application layer and parts of its transport layer.
- **Reasons:**
 - TCP/IP's transport protocols (e.g., TCP) include session-like functions.
 - Application layer protocols can incorporate presentation functions as needed.
- This integration makes TCP/IP more compact and practical for Internet use.

Figure 2.12 TCP/IP and OSI model



Lack of OSI Success: [more at end of table](#)

- **Established TCP/IP:** TCP/IP was widely adopted, and switching to OSI was **costly**.
- **Incomplete OSI Layers:** **Session** and **presentation** layers lacked fully defined protocols and software.
- **Performance Issues:** OSI implementations were **less efficient** than TCP/IP, discouraging adoption.

Additional Explanation: The OSI model provides a universal framework for understanding network communication, emphasizing modularity and standardization. While TCP/IP became the Internet's standard due to its practicality and early adoption, OSI remains valuable for educational purposes and designing new protocols. The absence of session and presentation layers in TCP/IP reflects a pragmatic design choice, embedding their functions in other layers to reduce complexity. Understanding OSI helps students appreciate the layered approach and its influence on modern networking standards.

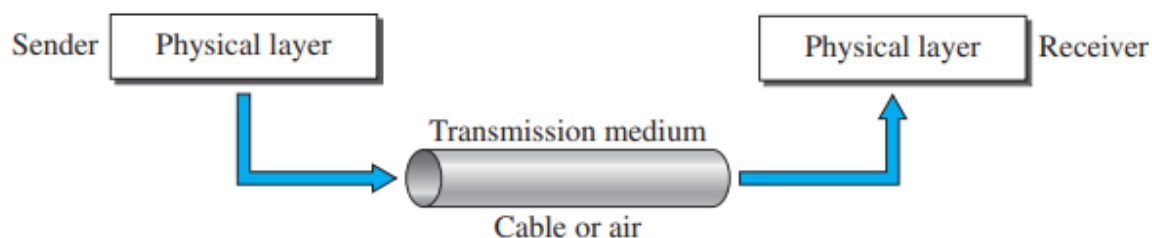
Aspect	OSI Model	TCP/IP Model
✓ Number of Layers	7 (Physical, Data Link, Network, Transport, Session, Presentation, Application)	4 (Network Access, Internet, Transport, Application)
Layer Mapping	- Physical + Data Link → Network Access	- Network Access (Physical + Data Link)
	- Network → Internet	- Internet (Network)
	- Transport → Transport	- Transport (Transport)

Aspect	OSI Model	TCP/IP Model
	- Session + Presentation + Application → Application	- Application (Session + Presentation + Application)
✓ Purpose	<u>Theoretical framework</u> for standardizing any network protocol.	<u>Practical framework</u> for internet-based communication.
✓ Granularity	<u>Highly</u> granular with <u>distinct</u> Session and Presentation layers.	<u>Less</u> granular, <u>merging</u> Session, Presentation, and Application for simplicity.
Protocols	Protocol-agnostic; uses concepts like X.25, TP0-TP4 (not widely adopted).	Specific protocols: TCP, UDP, IP, HTTP, DNS, ICMP, ARP.
✓ Adoption	Limited to education/reference; <u>rarely</u> implemented directly.	<u>Widely</u> implemented in the internet and real-world networks.
✓ Development	By <u>ISO</u> , slow committee-driven process.	By <u>DoD</u> for ARPANET, rapid and community-driven.
✓ Complexity	<u>Complex</u> due to seven layers and strict boundaries.	<u>Simpler</u> with four layers and overlapping functions.
✓ Flexibility	<u>Broadly</u> applicable to any network type.	Focused on IP-based networks, <u>less</u> flexible but practical.
Reasons for OSI's Failure	- Complex, over-abstracted structure.	
	- Lack of practical protocols.	
	- Slow standardization.	
	- Limited industry adoption.	
	- Poor timing vs. TCP/IP's rise.	
	- Inflexible for new technologies.	
	- Perceived as bureaucratic.	N/A (dominant due to simplicity, robust protocols, early adoption, adaptability).

. Transmission Media

Definition: Transmission media carry information between devices, located below the physical layer (layer zero). They are critical for determining the speed, distance, and reliability of data communication.

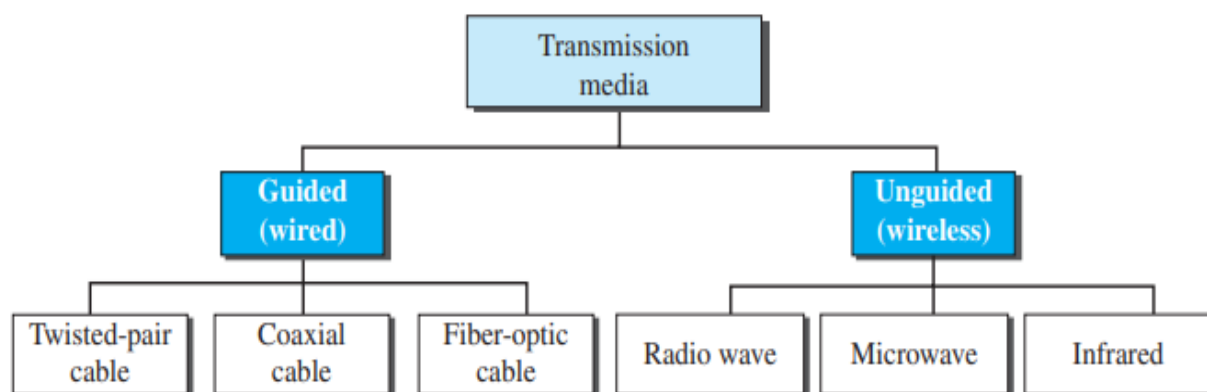
Figure 7.1 *Transmission medium and physical layer*



Categories (Figure 7.2):

1. **Guided Media:** Physical conduits (twisted-pair, coaxial, fiber-optic cables).
2. **Unguided Media:** Wireless signals (radio waves, microwaves, infrared).

Figure 7.2 *Classes of transmission media*



Guided Media: Guided Media: Guided media, which are those that provide a medium from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable.

1. **Twisted-Pair Cable** (Figure 7.3, 7.4):

- Twisted-Pair Cable: A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together. One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference.

- **Types:**

- **UTP:** Common, cost-effective, widely used.
- **STP:** Includes a metal shield for better noise protection, but bulkier and more expensive.

The most common UTP connector is RJ45 (RJ stands for registered jack)

Applications

- Twisted-pair cables are used in telephone lines to provide voice and data channels.

- Local-area networks, such as 10Base-T and 100Base-T, also use twisted-pair cables.
 - **Categories** (Table 7.1): EIA standards (CAT 1–7) define data rates and uses (e.g., CAT 5: 100 Mbps for LANs, CAT 7: 600 Mbps).
 - **Connector**: RJ45, a keyed connector ensuring correct insertion.
 - **Performance** (Figure 7.6): Attenuation increases sharply above 100 kHz, influenced by wire gauge (thicker wires have lower attenuation).
 - **Explanation**: Twisting balances interference, making UTP ideal for cost-sensitive applications like home networking. Categories reflect advancements in cable quality, enabling higher data rates for modern LANs.

Figure 7.3 *Twisted-pair cable*



Figure 7.4 *UTP and STP cables*

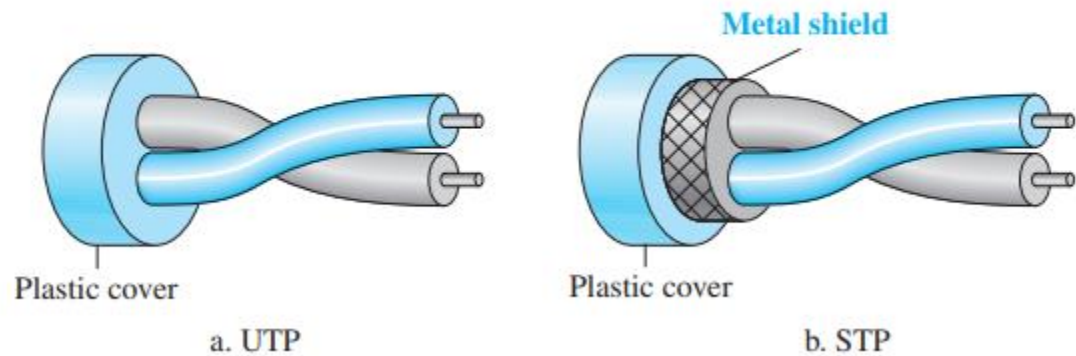


Table 7.1 *Categories of unshielded twisted-pair cables*

Category	Specification	Data Rate (Mbps)	Use
1	Unshielded twisted-pair used in telephone	< 0.1	Telephone
2	Unshielded twisted-pair originally used in T lines	2	T-1 lines
3	Improved CAT 2 used in LANs	10	LANs
4	Improved CAT 3 used in Token Ring networks	20	LANs
5	Cable wire is normally 24 AWG with a jacket and outside sheath	100	LANs

Category	Specification	Data Rate (Mbps)	Use
5E	An extension to category 5 that includes extra features to minimize the crosstalk and electromagnetic interference	125	LANs
6	A new category with matched components coming from the same manufacturer. The cable must be tested at a 200-Mbps data rate.	200	LANs
7	Sometimes called <i>SSTP (shielded screen twisted-pair)</i> . Each pair is individually wrapped in a helical metallic foil followed by a metallic foil shield in addition to the outside sheath. The shield decreases the effect of crosstalk and increases the data rate.	600	LANs

Figure 7.5 UTP connector

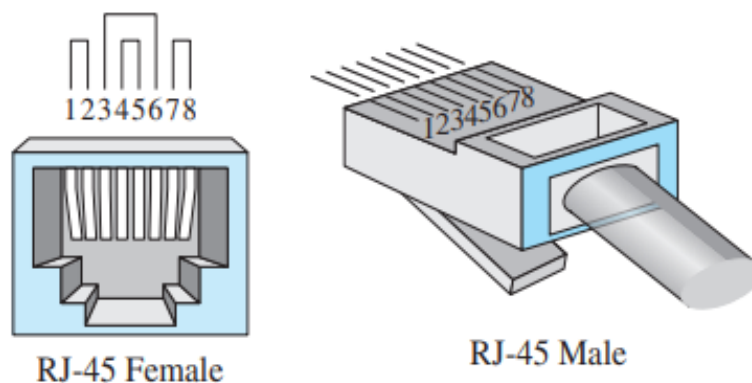
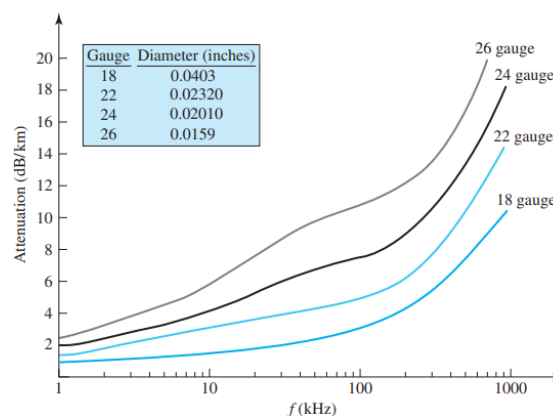


Figure 7.6 UTP performance



2. Coaxial Cable (Figure 7.7):

Coaxial cable (or *coax*) carries signals of higher frequency ranges than those in twisted pair cable. coax has a central core conductor of solid or stranded wire (usually copper) enclosed in an insulating sheath, which is, in turn, encased in an outer conductor of metal foil, braid, or a combination of the two. The outer

metallic wrapping serves both as a shield against noise and as the second conductor, which completes the circuit. This outer conductor is also enclosed in an insulating sheath, and the whole cable is protected by a plastic cover.

- The most common type of connector used today is the Bayone-Neill-Concelman (BNe), connector.

Applications

- Coaxial cable was widely used in analog telephone networks, digital telephone networks. Cable TV networks also use coaxial cables.
- Another common application of coaxial cable is in traditional Ethernet LANs
- **Explanation:** Coaxial cables offer higher bandwidth but are less common today due to fiber-optic advancements. Their shielding makes them robust against noise, suitable for early high-speed networks.

Figure 7.7 Coaxial cable

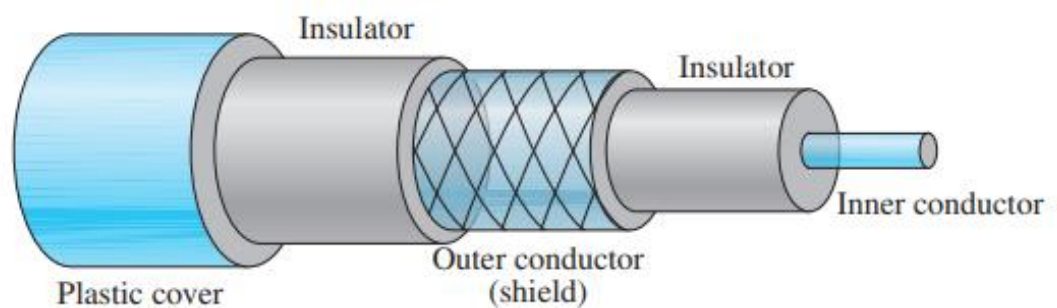
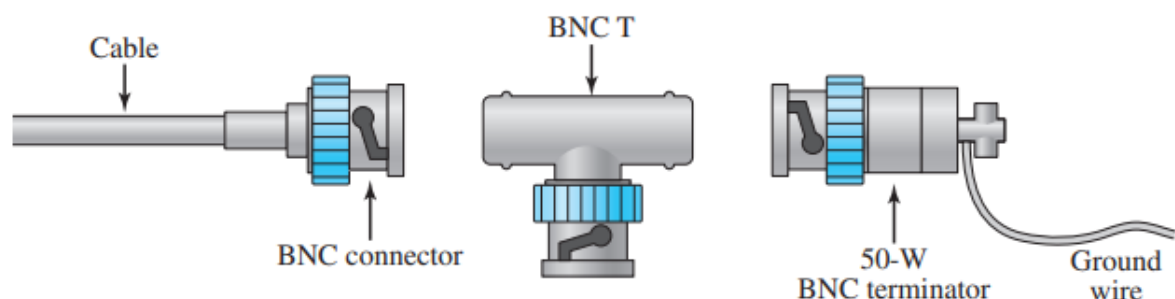


Table 7.2 Categories of coaxial cables

Category	Impedance	Use
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

Figure 7.8 BNC connectors



3. Fiber-Optic Cable (Figure 7.11, 7.14):

A fiber-optic cable is made of glass or plastic and transmits signals in the form of light. Light travels in a straight line as long as it is moving through a single uniform substance. If a ray of light traveling through one substance suddenly enters another substance (of a different density), the ray changes direction.

Bending of light ray

Optical fibers use reflection to guide light through a channel. A glass or plastic core is surrounded by a cladding of less dense glass or plastic.

Figure 7.11 Optical fiber

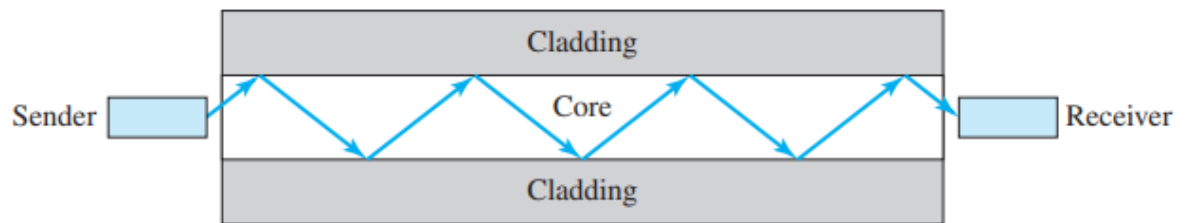
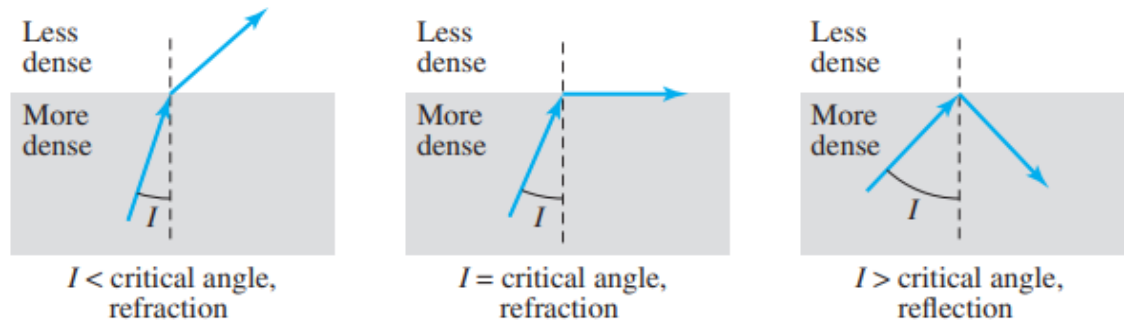
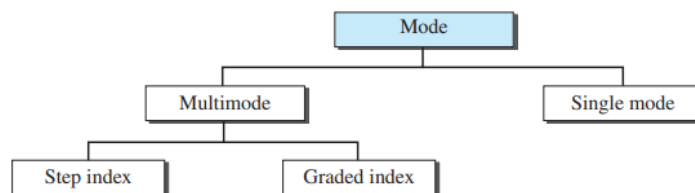


Figure 7.10 Bending of light ray



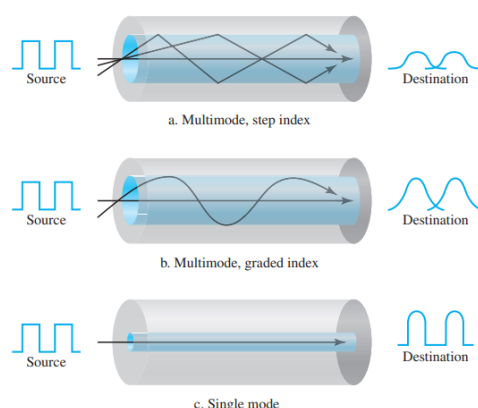
– Propagation Modes (Figure 7.12, 7.13):

Figure 7.12 Propagation modes



Multimode is so named because multiple beams from a light source move through the core in different paths. How these beams move within the cable depends on the structure of the core, as shown in Figure.

Figure 7.13 Modes



In **multimode step-index fiber**, the density of the core remains constant from the center to the edges. A beam of light moves through this constant density in a straight line until it reaches the interface of the core and the cladding. The term *step index* refers to the suddenness of this change, which contributes to the distortion of the signal as it passes through the fiber.

A second type of fiber, called **multimode graded-index fiber**, decreases this distortion of the signal through the cable. The word *index* here refers to the index of refraction.

Single-Mode: Single-mode uses step-index fiber and a highly focused source of light that limits beams to a small range of angles, all close to the horizontal.

Applications

- Fiber-optic cable is often found in backbone networks because its wide bandwidth is cost-effective..
- Some cable TV companies use a combination of optical fiber and coaxial cable, thus creating a hybrid network.
- Local-area networks such as 100Base-FX network (Fast Ethernet) and 1000Base-X also use fiber-optic cable

Advantages and Disadvantages of Optical Fiber

Advantages Fiber-optic cable has several advantages over metallic cable (twisted pair or coaxial).

1. Higher bandwidth.
2. Less signal attenuation. Fiber-optic transmission distance is significantly greater than that of other guided media. A signal can run for 50 km without requiring regeneration. We need repeaters every 5 km for coaxial or twisted-pair cable.
3. Immunity to electromagnetic interference. Electromagnetic noise cannot affect fiber-optic cables.
4. Resistance to corrosive materials. Glass is more resistant to corrosive materials than copper.
5. Light weight. Fiber-optic cables are much lighter than copper cables.

6. Greater immunity to tapping. Fiber-optic cables are more immune to tapping than copper cables. Copper cables create antenna effects that can easily be tapped.

Disadvantages There are some disadvantages in the use of optical fiber.

1. Installation and maintenance
2. Unidirectional light propagation. Propagation of light is unidirectional. If we need bidirectional communication, two fibers are needed.
3. Cost. The cable and the interfaces are relatively more expensive than those of other guided media. If the demand for bandwidth is not high, often the use of optical fiber cannot be justified.

Explanation: Fiber-optic cables are ideal for high-speed, long-distance communication due to their superior bandwidth and reliability. Their use in backbones and high-speed LANs reflects their ability to handle massive data volumes, though cost and expertise requirements limit their use in low-demand scenarios.

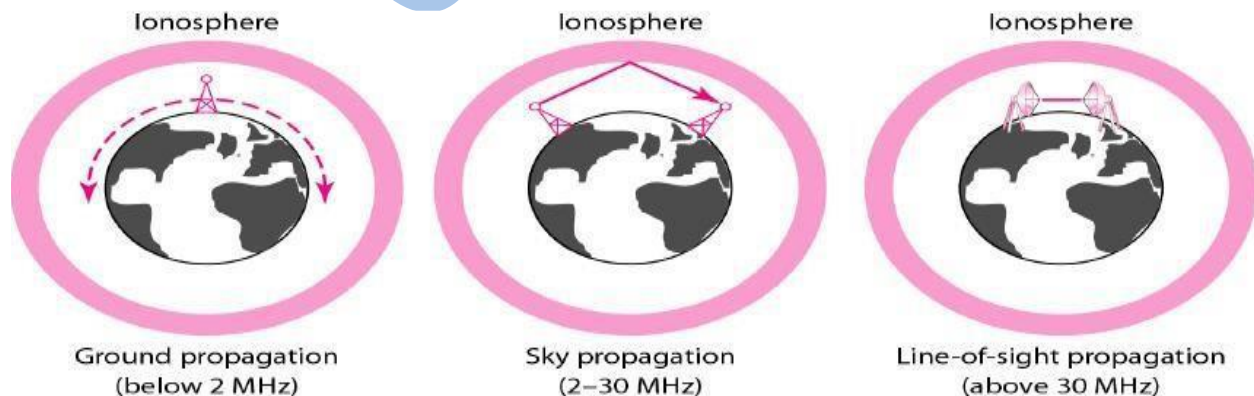
UNGUIDED MEDIA: WIRELESS

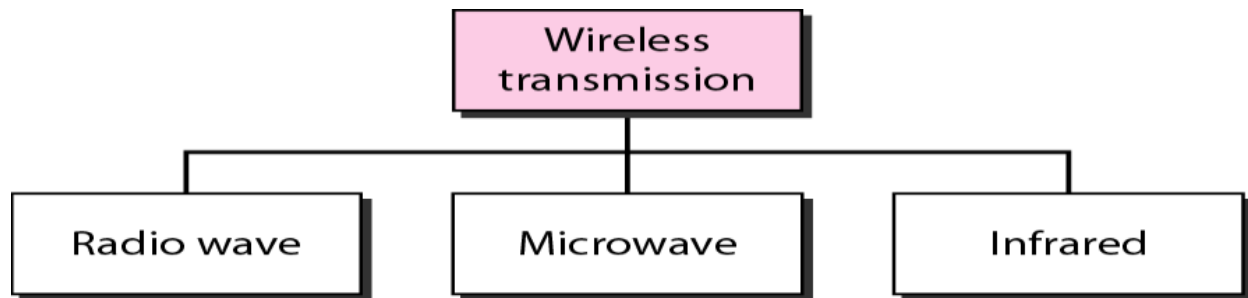
Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication.

1. Radio Waves
2. Microwaves
3. Infrared



Unguided signals can travel from the source to destination in several ways: ground propagation, sky propagation, and line-of-sight propagation, as shown in Figure



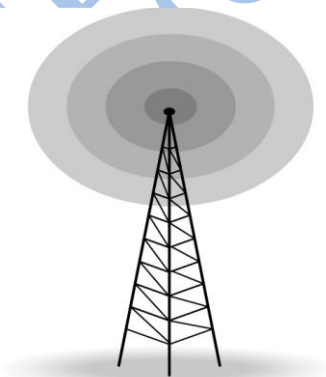


Radio Waves

Electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are normally called radio waves. Radio waves are omni directional. When an antenna transmits radio waves, they are propagated in all directions. This means that the sending and receiving antennas do not have to be aligned. A sending antenna sends waves that can be received by any receiving antenna. The omni directional property has a disadvantage, too. The radio waves transmitted by one antenna are susceptible to interference by another antenna that may send signals using the same frequency or band.

Omni directional Antenna

Radio waves use omnidirectional antennas that send out signals in all directions. Based on the wavelength, strength, and the purpose of transmission, we can have several types of antennas. Figure shows an omnidirectional antenna.



Applications

The Omni directional characteristics of radio waves make them useful for multicasting, in which there is one sender but many receivers. AM and FM radio, television, maritime radio, cordless phones, and paging are examples of multicasting.

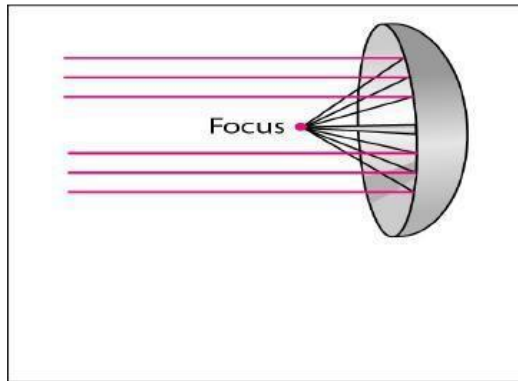
Microwaves

Electromagnetic waves having frequencies between 1 and 300 GHz are called microwaves. Microwaves are unidirectional. The sending and receiving antennas need to be aligned. The

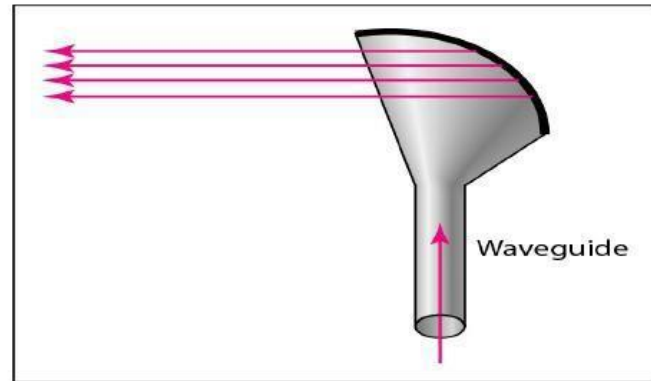
unidirectional property has an obvious advantage. A pair of antennas can be aligned without interfering with another pair of aligned antennas

Unidirectional Antenna

Microwaves need unidirectional antennas that send out signals in one direction. Two types of antennas are used for microwave communications: the parabolic dish and the horn



a. Dish antenna



b. Horn antenna

Applications:

Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LANs

Infrared

Infrared waves, with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm), can be used for short-range communication. Infrared waves, having high frequencies, cannot penetrate walls. This advantageous characteristic prevents interference between one system and another; a short-range communication system in one room cannot be affected by another system in the next room.

When we use our infrared remote control, we do not interfere with the use of the remote by our neighbors. Infrared signals are useless for long-range communication. In addition, we cannot use infrared waves outside a building because the sun's rays contain infrared waves that can interfere with the communication.

Applications:

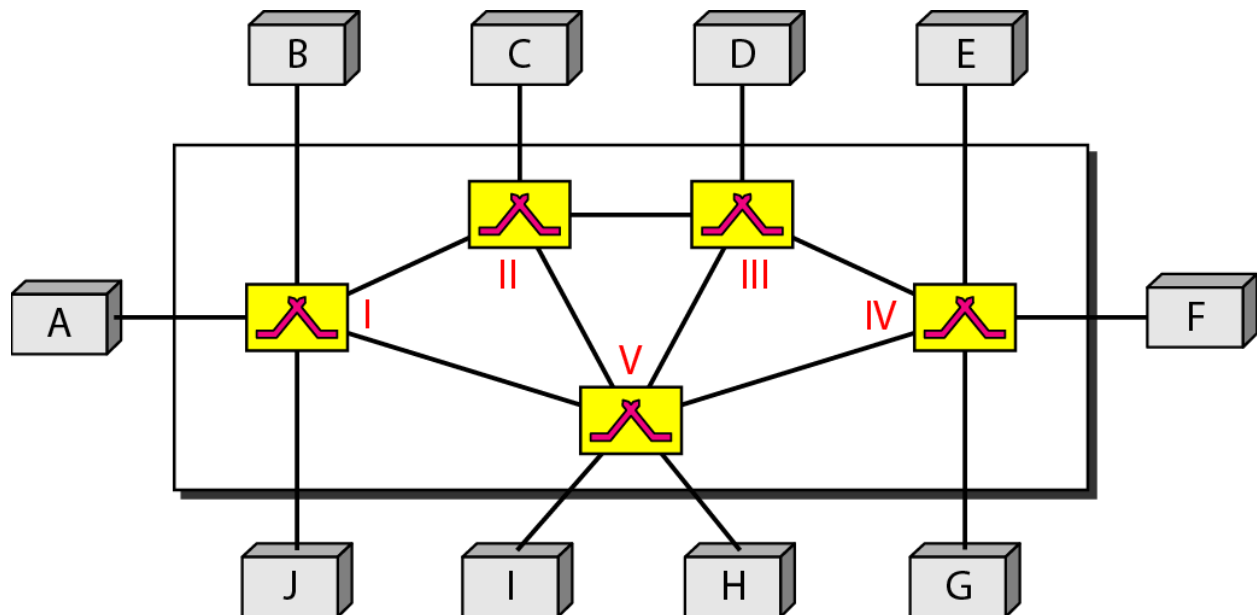
Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.

Switching

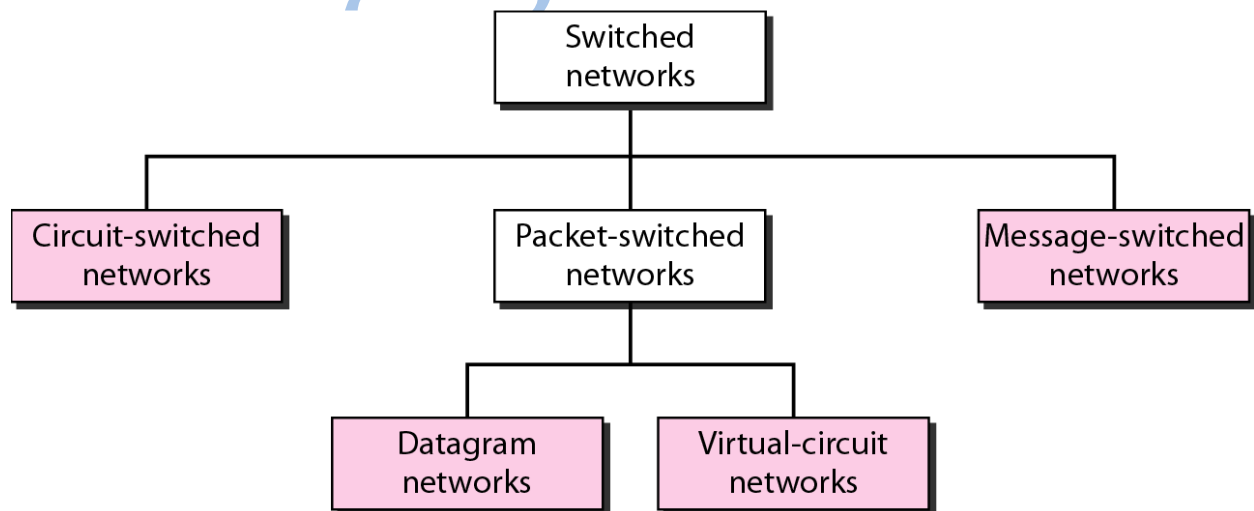
A network is a set of connected devices. Whenever we have multiple devices, we have the problem of how to connect them to make one-to-one communication possible. One solution is to make a point-to-point connection between each pair of devices (a mesh topology) or between a central device and every other device (a star topology). These methods, however, are impractical and wasteful when applied to very large networks.

The number and length of the links require too much infrastructure to be cost-efficient, and the majority of those links would be idle most of the time.

A better solution is switching. A switched network consists of a series of interlinked nodes, called switches. Switches are devices capable of creating temporary connections between two or more devices linked to the switch. In a switched network, some of these nodes are connected to the end systems (computers or telephones, for example). Others are used only for routing. Figure shows a switched network.



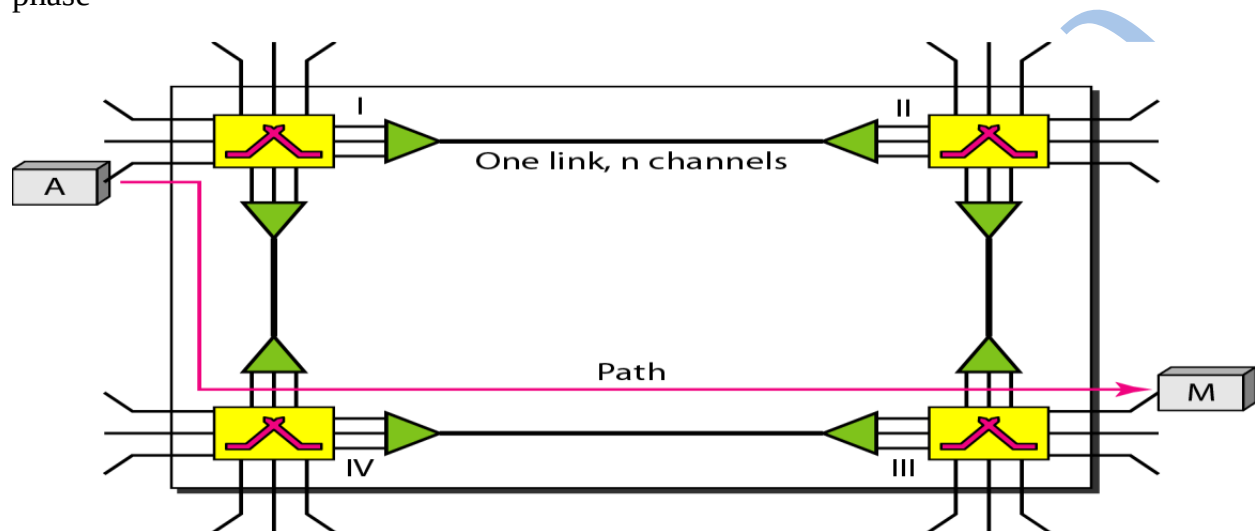
We can then divide today's networks into three broad categories: circuit-switched networks, packet-switched networks, and message-switched. Packet-switched networks can further be divided into two subcategories-virtual-circuit networks and datagram networks as shown in Figure.



CIRCUIT-SWITCHED NETWORKS

A circuit-switched network consists of a set of switches connected by physical links. A connection between two stations is a dedicated path made of one or more links. However, each connection uses only one dedicated channel on each link. Each link is normally divided into n channels by using FDM or TDM.

In circuit switching, the resources need to be reserved during the setup phase; the resources remain dedicated for the entire duration of data transfer until the teardown phase



Three Phases

The actual communication in a circuit-switched network requires three phases: connection setup, data transfer, and connection teardown.

Setup Phase

Before the two parties (or multiple parties in a conference call) can communicate, a dedicated circuit (combination of channels in links) needs to be established. Connection setup means creating dedicated channels between the switches. For example, in Figure, when system A needs to connect to system M, it sends a setup request that includes the address of system M, to switch I. Switch I finds a channel between itself and switch IV that can be dedicated for this purpose. Switch I then sends the request to switch IV, which finds a dedicated channel between itself and switch III. Switch III informs system M of system A's intention at this time.

In the next step to making a connection, an acknowledgment from system M needs to be sent in the opposite direction to system A. Only after system A receives this acknowledgment is the connection established.

Data Transfer Phase

After the establishment of the dedicated circuit (channels), the two parties can transfer data.

Teardown Phase

When one of the parties needs to disconnect, a signal is sent to each switch to release the resources.

Efficiency

It can be argued that circuit-switched networks are not as efficient as the other two types of networks because resources are allocated during the entire duration of the connection. These resources are unavailable to other connections.

Delay

Although a circuit-switched network normally has low efficiency, the delay in this type of network is minimal. During data transfer the data are not delayed at each switch; the resources are allocated for the duration of the connection.

The total delay is due to the time needed to create the connection, transfer data, and disconnect the circuit.

Switching at the physical layer in the traditional telephone network uses the circuit-switching approach.

DATAGRAM NETWORKS

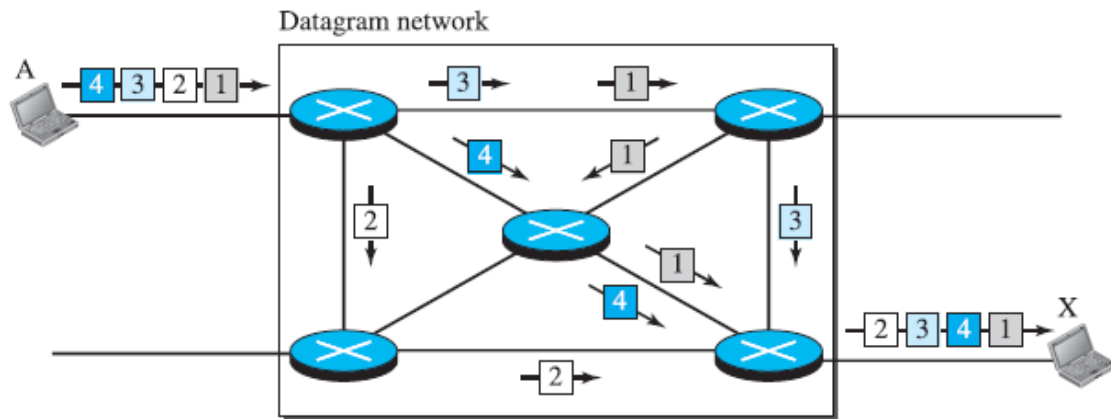
In a packet-switched network, there is no resource reservation; resources are allocated on demand. The allocation is done on a first come, first-served basis. When a switch receives a packet, no matter what is the source or destination, the packet must wait if there are other packets being processed. This lack of reservation may create delay. For example, if we do not have a reservation at a restaurant, we might have to wait.

In a datagram network, each packet is treated independently of all others. Packets in this approach are referred to as datagrams. Datagram switching is normally done at the network layer.

Figure shows how the datagram approach is used to deliver four packets from station A to station X. The switches in a datagram network are traditionally referred to as routers.

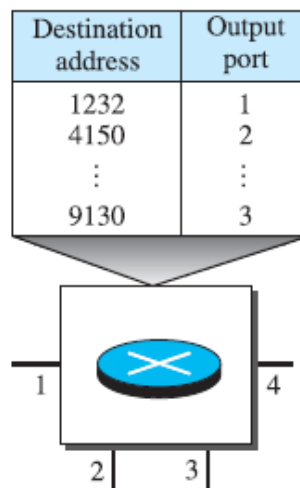
The datagram networks are sometimes referred to as connectionless networks. The term **connectionless** here means that the switch (packet switch) does not keep information about the connection state. There are no setup or teardown phases. Each packet is treated the same by a switch regardless of its source or destination.

Figure 8.7 A datagram network with four switches (routers)



A switch in a datagram network uses a routing table that is based on the destination address. The destination address in the header of a packet in a datagram network remains the same during the entire journey of the packet.

Figure 8.8 Routing table in a datagram network



Efficiency

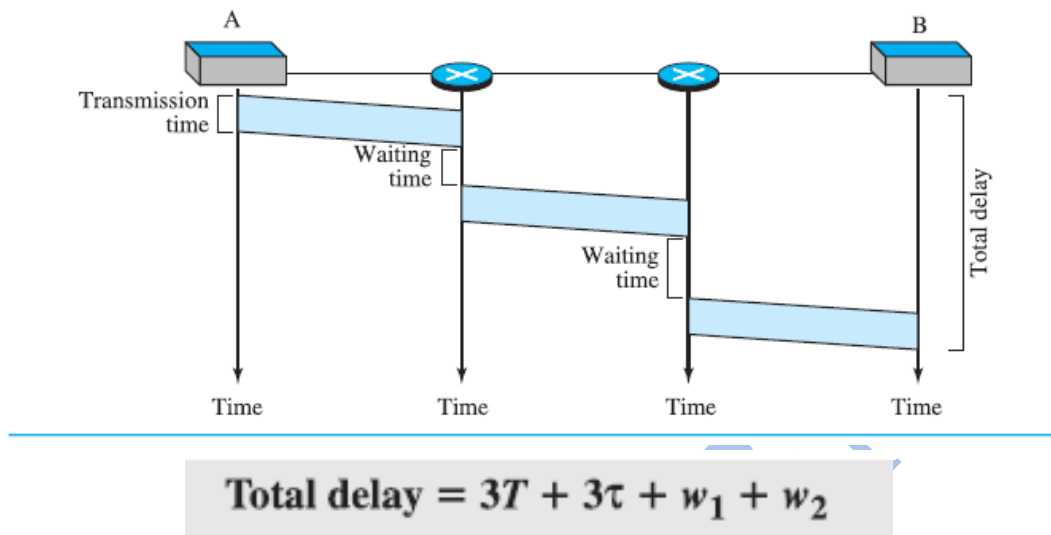
The efficiency of a datagram network is better than that of a circuit-switched network; resources are allocated only when there are packets to be transferred.

Delay

There may be greater delay in a datagram network than in a virtual-circuit network. Although there are no setup and teardown phases, each packet may experience a wait at a switch before it is forwarded. In addition, since not all packets in a message necessarily travel through the same switches, the delay is

not uniform for the packets of a message.

Figure 8.9 Delay in a datagram network



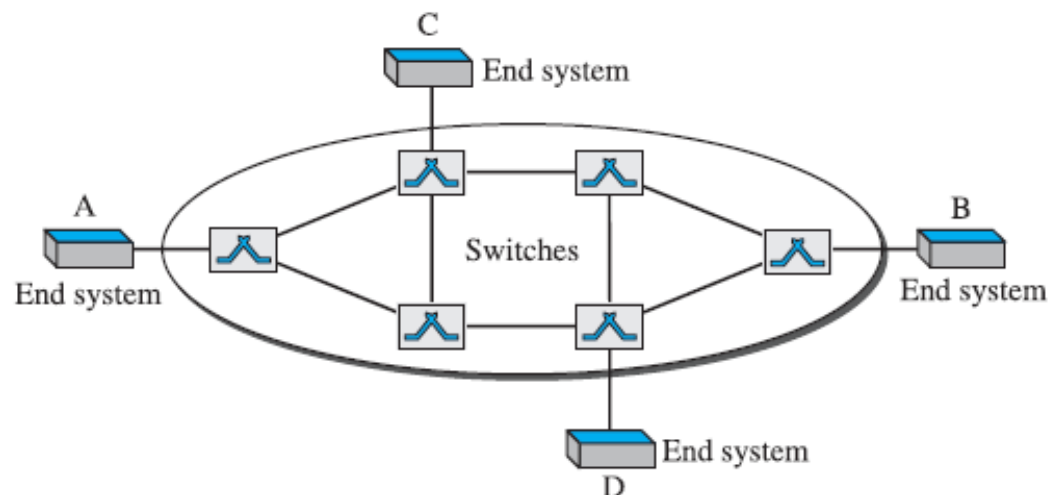
Switching in the Internet is done by using the datagram approach to packet switching at the network layer.

GPT

VIRTUAL-CIRCUIT NETWORKS

A virtual-circuit network is a cross between a circuit-switched network and a datagram network. It has some characteristics of both.

Figure 8.10 Virtual-circuit network



1. As in a circuit-switched network, there are setup and teardown phases in addition to the data transfer phase.
2. Resources can be allocated during the setup phase, as in a circuit-switched network, or on demand, as in a datagram network.
3. As in a datagram network, data are packetized and each packet carries an address in the header. However, the address in the header has local jurisdiction (it defines what should be the next switch and the channel on which the packet is being carried), not end-to-end jurisdiction.
4. As in a circuit-switched network, all packets follow the same path established during the connection.
5. A virtual-circuit network is normally implemented in the data link layer, while a circuit-switched network is implemented in the physical layer and a datagram network in the network layer.

Addressing

In a virtual-circuit network, two types of addressing are involved: global and local (virtual-circuit identifier).

Global Addressing

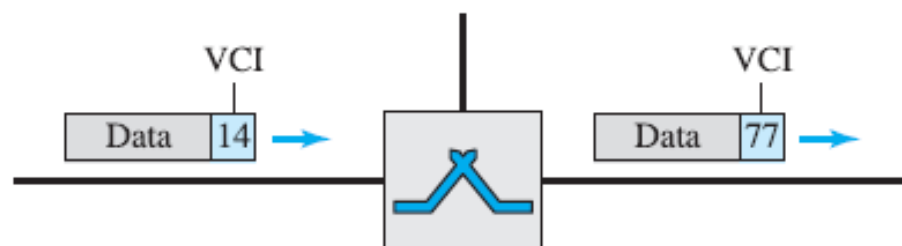
A source or a destination needs to have a global address—an address that can be unique in the scope of the network.

Virtual-Circuit Identifier

The identifier that is actually used for data transfer is called the virtual-circuit identifier (VCI). A VCI, unlike a global address, is a small number that has only switch scope; it is used by a frame between two switches. When a frame arrives at a switch, it has a VCI; when it leaves, it has a different VCI.

Figure shows how the VCI in a data frame changes from one switch to another. Note that a VCI does not need to be a large number since each switch can use its own unique set of VCIs.

Figure 8.11 *Virtual-circuit identifier*

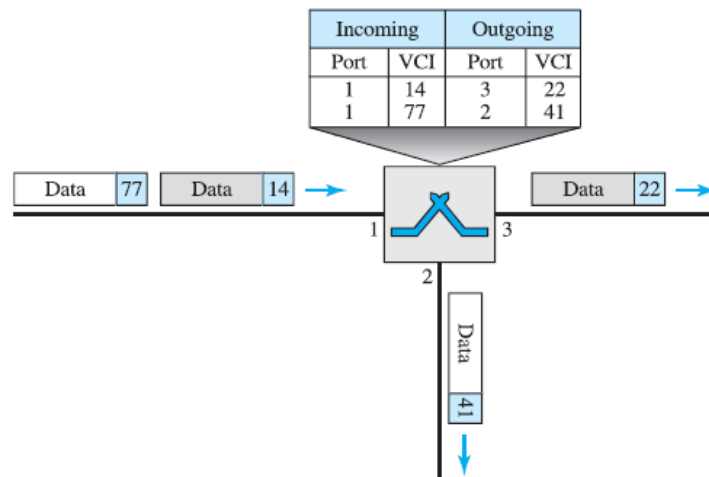


Three Phases

Three phases in a virtual-circuit network: setup, data transfer, and teardown. We first discuss the data transfer phase, which is more straightforward; we then talk about the setup and teardown phases.

Data Transfer Phase

Figure 8.12 Switch and tables in a virtual-circuit network



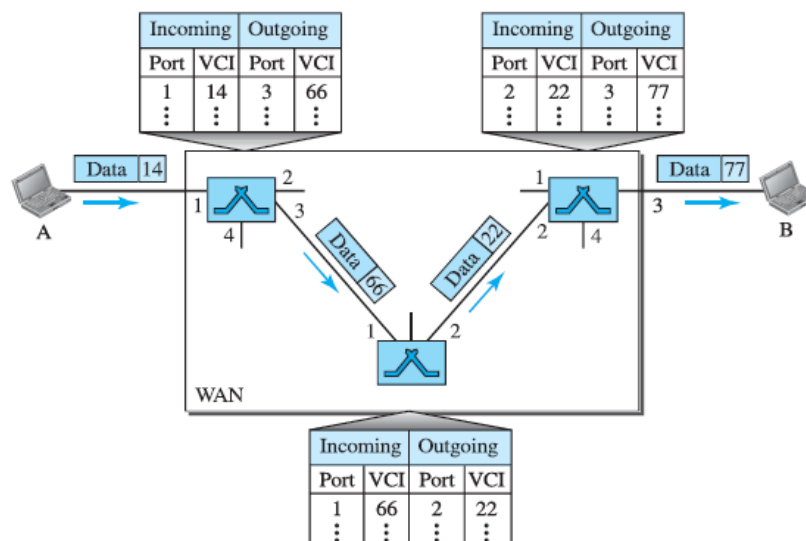
To transfer a frame from a source to its destination, all switches need to have a table entry for this virtual circuit. The table, in its simplest form, has four columns.

We show later how the switches make their table entries, but for the moment we assume that each switch has a table with entries for all active virtual circuits. Figure shows such a switch and its corresponding table.

Figure shows a frame arriving at port 1 with a VCI of 14. When the frame arrives, the switch looks in its table to find port 1 and a VCI of 14. When it is found, the switch knows to change the VCI to 22 and send out the frame from port 3.

Figure shows how a frame from source A reaches destination B and how its VCI changes during the trip.

Figure 8.13 Source-to-destination data transfer in a virtual-circuit network



Each switch changes the VCI and routes the frame.

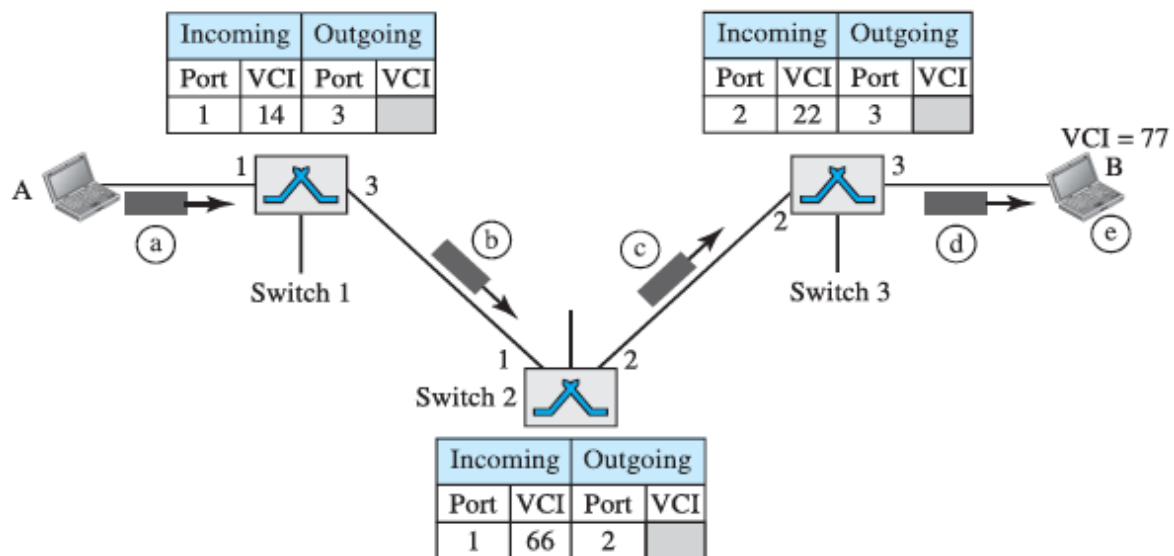
The data transfer phase is active until the source sends all its frames to the destination. The procedure at the switch is the same for each frame of a message. The process creates a virtual circuit, not a real circuit, between the source and destination.

Setup Phase

In the setup phase, a switch creates an entry for a virtual circuit. For example, suppose source A needs to create a virtual circuit to B. Two steps are required: the setup request and the acknowledgment.

Setup Request A setup request frame is sent from the source to the destination. Figure shows the process.

Figure 8.14 Setup request in a virtual-circuit network



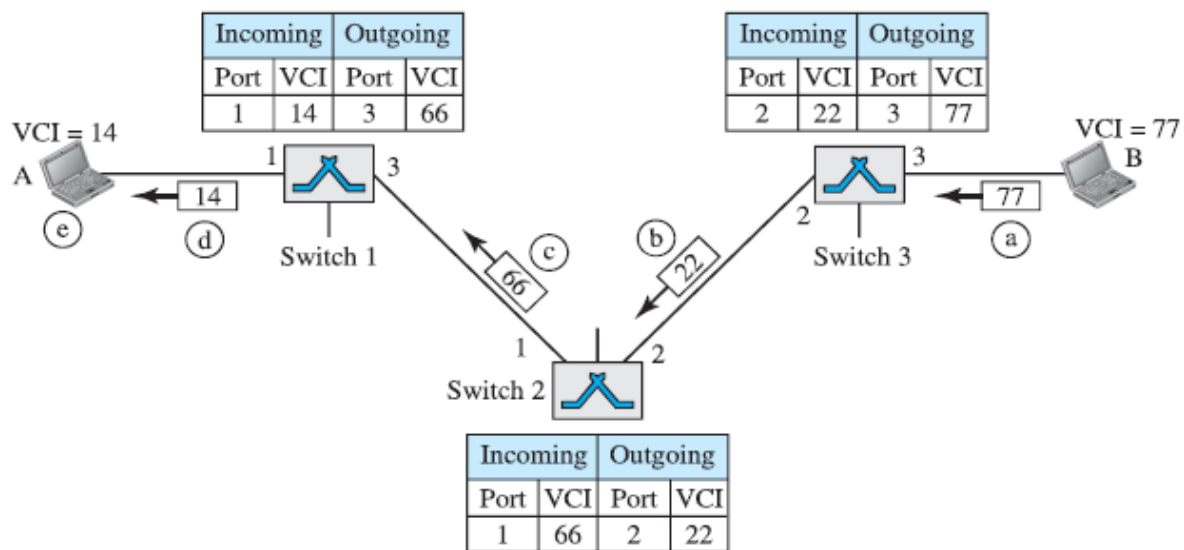
1. Source A sends a setup frame to switch 1.
2. Switch 1 receives the setup request frame. It knows that a frame going from A to B goes out through port 3. For the moment, assume that it knows the output port. The switch creates an entry in its table for this virtual circuit, but it is only able to fill three of the four columns. The switch assigns the incoming port (1) and chooses an available incoming VCI (14) and the outgoing port (3). It does not yet know the outgoing VCI, which will be found during the acknowledgment step. The switch then forwards the frame through port 3 to switch 2.
3. Switch 2 receives the setup request frame. The same events happen here as at switch 1; three columns of the table are completed: in this case, incoming port (1), incoming VCI (66), and outgoing port (2).
4. Switch 3 receives the setup request frame. Again, three columns are completed: incoming port (2), incoming VCI (22), and outgoing port (3).

- Destination B receives the setup frame, and if it is ready to receive frames from A, it assigns a VCI to the incoming frames that come from A, in this case 77. This VCI lets the destination know that the frames come from A, and not other sources.

Acknowledgment A special frame, called the acknowledgment frame, completes the entries in the switching tables.

Figure shows the process.

Figure 8.15 Setup acknowledgment in a virtual-circuit network



The destination sends an acknowledgment to switch 3. The acknowledgment carries the global source and destination addresses so the switch knows which entry in the table is to be completed. The frame also carries VCI 77, chosen by the destination as the incoming VCI for frames from

- Switch 3 uses this VCI to complete the outgoing VCI column for this entry. Note that 77 is the incoming VCI for destination B, but the outgoing VCI for switch 3.
- Switch 3 sends an acknowledgment to switch 2 that contains its incoming VCI in the table, chosen in the previous step. Switch 2 uses this as the outgoing VCI in the table.
- Switch 2 sends an acknowledgment to switch 1 that contains its incoming VCI in the table, chosen in the previous step. Switch 1 uses this as the outgoing VCI in the table.
- Finally switch 1 sends an acknowledgment to source A that contains its incoming VCI in the table, chosen in the previous step.
- The source uses this as the outgoing VCI for the data frames to be sent to destination B.

Teardown Phase

In this phase, source A, after sending all frames to B, sends a special frame called a *teardown request*. Destination B responds with a teardown confirmation frame. All switches delete the corresponding entry from their tables.

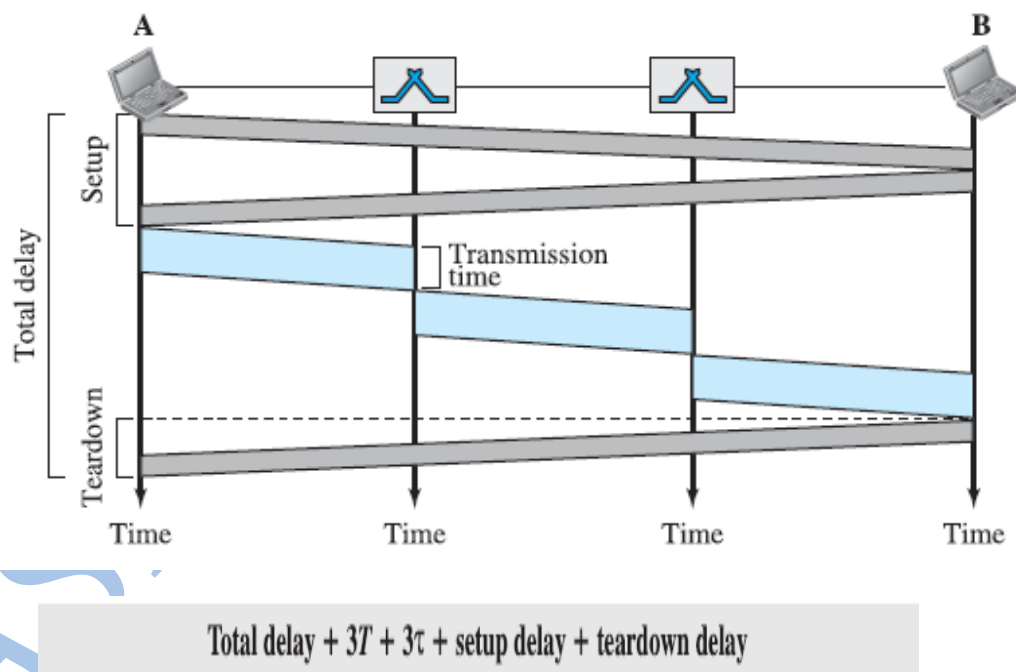
Efficiency

In virtual-circuit switching, all packets belonging to the same source and destination travel the same path; but the packets may arrive at the destination with different delays if resource allocation is on demand.

Delay

In a virtual-circuit network, there is a one-time delay for setup and a one-time delay for teardown. If resources are allocated during the setup phase, there is no wait time for individual packets. Figure shows the delay for a packet traveling through two switches in a virtual-circuit network

Figure 8.16 Delay in a virtual-circuit network



Switching at the data link layer in a switched WAN is normally implemented by using virtual-circuit techniques.

Comparison

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC